All-Weather IR-Based Smart Vehicle Gate Recognition System

Final Year Project Report

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Lahore Campus

July 2026 *Declaration*

We, hereby declare that this project neither as a whole nor as a part there of has been copied out from any source. It is further declared that we have developed this project and the accompanied report entirely on the basis of our personal efforts made under the sincere guidance of our supervisor. No portion of the work presented in this report has been submitted in the support of any other degree or qualification of this or any other University or Institute of learning, if found we shall stand responsible.

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COMSATS UNIVERSITY ISLAMABAD,
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An Undergraduate Final Year Project Report submitted to the Department of

COMPUTER ENGINEERING

As a Partial Fulfillment for the award of Degree

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by

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No portion of the work referred in this report has been submitted in fulfilment of another degree or qualification for any other institute or university.

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Table of Contents

1 INTRODUCTION 1

1.1 Historical Background 1 1.2
Objectives and Limitations 2 1.3
Project Scope 3 1.4
Assumptions and Dependencies 4 1.5
Broader Impact (UN SDGs) 4 1.6
Thesis Key Contributions 6
1.6.1 Thesis Outline 7

2 LITERATURE REVIEW 9

2.1 RELATEDTECHNOLOGIES 9
2.1.1 *ObjectDetection 9*
2.1.2 *Ocr 10*
2.1.3 *ImagePreprocessing 10*
2.1.4 *Model Recognition 11*
2.2 OverviewofexisistingSystems 11
2.2.1 *AlprSystems 12*
2.2.2 *AlprResearch 12*
2.3 LIMITATIONS 13

3 MODELING AND SIMULATION 15

3.1 ConceptualFramework 15 3.2
High Level Design 15
3.2.1 Block Diagram 15
3.2.2 Flow Diagram 16
3.2.3 Performance 17
3.3 Subsystem Design 18
3.1.1 *ListofSubsystems 18*
3.1.2 *Flowchart 19*
3.1.3 3.4 LowLevelDesignandComponentSelection 19
3.4.1ImageAcquisitio 20
3.4.2LicensePlateDetection 20
3.4.3ImagePreprocessing 20
3.4.4OpticalCharacterRecognition 20
3.4.5 Vehicle Classification 20
3.4.6 Authorization and Decision 21
3.4.7 EmbeddedProcessing 21
3.4.8 User Interface 21
3.5 SYSTEMARCHITECTURE 22

4 HARDWARE/SOFTWARE IMPLEMENTATION AND TESTING 27

4.1 SOFTWARE IMPLEMENTATION 27
4.1.1 *LicensePlateDetection 27*
4.1.2 *AdaptiveImagePrrprocessing 27*

4.1.3 *TextRecognition* *28*
4.1.4 *VehcileMakeandModelClassifications* *28*
3. 1.5AuthorizationLogic *28*
4.1.6 Visualization and Logic *28*
4.2 TESTING PROCEDURES 28
4.2.1 *DatasetDescription* *29*
4.2.2 *TestingMethodology* *29*
4.3 RESULTS AND FINDINGS 30
4.3.1 *YoloPlateDetectionResults* *30*
4.3.2 *PreprocessingandYOLOV8Results* *32*
4.3.3 *VehicleModelClassificationResults* *33*
4.3.4 *PipelineResults* *35*
4.4 WEB DASHBOARD IMPLEMENTATION 37
CHAPTER 05 48
Results and Discussion 49
5.1 License Plate Detection Results 50
5.1.1 Training Configuration 50
5.1.2 DetectionPerformance 50
5.2 Character Recognition Results 52
5.2.1 Dataset Details 52
CHAPTER 06 65
Conclusion and Future Work 66
6.1 Summary 67
6.2 Key Achievements 67
6.3 System Performance Summary 68
6.4 Limitations 68
6.5 Future Work 68
6.6 Conclusion 68
APPENDIXES 38
Appendix A: Sustainable Development Goals Achievement *38*
Appendix B: HDL or C Source Code *41*
APPENDIX C: LIST OF COMPONENTS 44
APPENDIX D: PROJECT TIMELINE 46
BIBLIOGRAPHY 47

List of Symbols, Abbreviations and

Acronyms

ALPR ..... Automatic License Plate Recognition

VMMR ..... Vehicle Make and Model Recognition

YOLO ..... You Only Look Once

OCR ..... Optical Character Recognition

List of Figures

Figure 3. 2. 1 Block Diagram 16

Figure 3. 2. 2 Flow Diagram 17
Figure 3. 3. 2 Flow Diagram of Subsystem 19
Figure 3. 4. 1 Prototype Application Diagram 22
Figure 3. 5. 1 Architecture Overview Diagram 23
Figure 3. 5. 1 Class Diagram 24
Figure 3. 5. 2 Erd Diagram 25

Figure 4. 3. 1 Performance Metrices HeatMap 30

Figure 4. 3. 1 Figure-2 Yolo Number Plate Detection 31
Figure 4. 3. 1 Figure-3 Confusion Matrix 31
Figure 4. 3. 2 Number Plate Preprocessing 32
Figure 4. 3. 2 Figure-2 Number Plate Recognition 33
Figure 4. 3. 3 Vehicle Make/Model Heatmap 33
Figure 4. 3. 3 Figure-2 Efficient-Net Confusion Matrix 34
Figure 4. 3. 3 Figure-3 Testing Example..... 34
Figure 4. 3. 4 Pipeline Result 1..... 35
Figure 4. 3. 4 Pipeline Result 2..... 36
Figure 4. 4. 1 Web Application Frontend 37

Figure 5.1 YOLOv8 License Plate Detection — Training and Validation Metrics Over 50 Epochs ..... ……..51
Figure 5.2 YOLOv8 License Plate Detection — Sample Detection on Pakistani Vehicle 51
Figure 5.3 YOLOv8 License Plate Detection — Confusion Matrix 52
Figure 5.4 Character Recognition Dataset — Roboflow Interface Showing 1,235 Annotated Plate Images …….. 53
Figure 5.5 Character Recognition — Training and Validation Metrics Over 100 Epochs 54
Figure 5.6 Character Recognition — Confusion Matrix (38 Classes) 55
Figure 5.7 Character Recognition — End-to-End Plate Detection and yolov8n Results on Test Images ... ……..56
Figure 5.8 VMMR — Training and Validation Loss, Accuracy, and Learning Rate Over 50 Epochs 58
Figure 5.9 VMMR — Test Metrics Heatmap (Accuracy, Precision, Recall, F1, FPR)

..... 59
Figure 5.10 VMMR — Confusion Matrix (Classes 0–49) 59
Figure 5.11 VMMR — Sample Test Predictions with Top-5 Probabilities 60
Figure 5.12 Original Foggy Vehicle Images 61
Figure 5.1 YOLOv8 License Plate Detection — Training and Validation Metrics Over 50 Epochs ..... ……..51
Figure 5.2 YOLOv8 License Plate Detection — Sample Detection on Pakistani Vehicle 51
Figure 5.3 YOLOv8 License Plate Detection — Confusion Matrix 52
Figure 5.4 Character Recognition Dataset — Roboflow Interface Showing 1,235 Annotated Plate Images …… …….. 53
Figure 5.5 Character Recognition — Training and Validation Metrics Over 100 Epochs 54
Figure 5.6 Character Recognition — Confusion Matrix (38 Classes) 55
Figure 5.7 Character Recognition — End-to-End Plate Detection and yolov8n Results on Test Images ... ……..56
Figure 5.8 VMMR — Training and Validation Loss, Accuracy, and Learning Rate Over 50 Epochs 58
Figure 5.9 VMMR — Test Metrics Heatmap (Accuracy, Precision, Recall, F1, FPR) 59
Figure 5.10 VMMR — Confusion Matrix (Classes 0–49) 59
Figure 5.11 VMMR — Sample Test Predictions with Top-5 Probabilities 60
Figure 5.12 Original Foggy Vehicle Images 61
Figure 5.13 After Fog Removal (Preprocessed) 61
Figure 5.14 Original Rainy Vehicle Images 62
Figure 5.15 After Rain Removal (Preprocessed) 62
Figure 5.16 Integrated Pipeline Results on Fog-Preprocessed Images — Plate Detection, yolov8n, and Vehicle Classification …….63
Figure 5.17 Integrated Pipeline Results on Rain-Preprocessed Images — Plate Detection, yolov8n, and Vehicle Classification …….. 63

Chapter 6 — No figures numbered in this document (use Table 6.1 for the performance summary in 6.3).

Appendix E Gantt Chart .....
46

List of Tables

Table 1. 1 SDGs in the light of objectives..... 5

Table 2. 3 Literature Review Summary Table 13

Table 4. 3 Result Table 36

Table 5.1 — License Plate Detection.....37

Table 5.2---Performance Matrix.....37

Table 5.3 — Dataset Summary 40

Table 5.4 —Training Configuration 40

Table 5.5 — Performance Metrics 41

Table 5.6 — Dataset Summary 43

Table 5.7 — Training Configuration 44

Table 5.8 — Test Set Performance Metrics 44

Table 5.9 — Training Progression 45

Abstract

This project presents a Smart Vehicle Gate Recognition System designed to automate vehicle access decisions in gated communities and parking facilities while reducing reliance on manual checks. The system integrates YOLO-based license plate detection, optical character recognition (OCR), and convolutional neural networks for vehicle make and model classification. The pipeline of image preprocessing is used to improve the lowlight images, decrease the noise, diminishes the fog and glare and refines the edges of characters to make YOLO V8 more strong in the real scenarios. The classification of vehicles is achieved on the basis of a fine-tuned EfficientNet-B0 model, and the ResNet18 is applied to benchmarking based on transfer learning. The decision to grant an authorization is made by comparing known license plates and vehicle models projected in an authorized database and the visualization of results is implemented on the basis of a web-based dashboard of dynamic control and administration. In the work of the future, physical gate control is also to be provided.

CHAPTER 01 — Introduction

1. Introduction

Vehicle identification is a basic necessity of a controlled access environment like a parking place, residential complex, and safe institutional premise. The traditional vehicle gate systems are usually based on manual checks or physical credentials that are not very efficient, have high chances of abuse, and cannot be easily operated on a large scale. The latest advances in computer vision and deep learning have opened the possibility of automated visual classification of vehicles through simple camera devices, which represents a beneficial alternative to access control based on credentials.

This Final Year Project (FYP-1) is designed to test a hybrid recognition pipeline that uses license plate detection, optical character recognition, and vehicle make and model recognition to enhance access control. The system is intended to perform efficiently under real-life conditions such as low light, glare, and poor weather conditions while making correct and consistent decisions. The design is scalable and modifiable to fit various facilities; the software-based design will enable support for hardware gates and centralized monitoring systems in the future.

1.1 Historical Background

The creation of intelligent vehicle gate systems has received growing attention in recent years due to rises in artificial intelligence and computer vision. Conventional gate entry systems — such as manual checking, RFID tags, or barcodes — are usually slow, subject to errors, and insecure in the event of losing or damaging credentials. They also struggle to handle crowded areas such as parking lots, corporate campuses, toll plazas, and other secure facilities.

Modern smart gate systems address these shortcomings by integrating automated licence plate recognition with deep learning, allowing contactless identification and authorisation in real time. Object detection networks like YOLO and yolov8n tools like EasyOCR can accurately identify license plates and extract plate text. Convolutional neural networks for

vehicle make and model recognition add an additional security layer, ensuring the recognized plate matches the actual vehicle.

1.2 Objectives and Limitations

The central purpose of this project is to explore automated vehicle access control based on computer vision and deep learning. The essential purposes are:

- Create a real-time YOLO-based license plate detection unit to identify and locate license plates in vehicle images.
- Write an intelligent YOLOv8 pipeline with EasyOCR and pre-processing options including denoising, histogram equalization, CLAHE, detail enhancement, and sharpening.
- Detect vehicle types using EfficientNet-B0 for make and model classification.
- Install authorization logic that cross-checks identified license plate text and vehicle model against an authorized vehicles database.
- Test performance in various conditions (day/night, glare, fog, rain, motion blur) targeting above 90% accuracy.

Limitations:

- This phase involves software implementation only; no real-time video streaming hardware or physical gates are offered.
- System performance relies on the quality of camera images and can degrade under extreme weather or very low light.
- Hardware integration including IR cameras and gate actuation is planned for future project stages.

1.3 Project Scope

This FYP-1 is dedicated to the software portion of the smart vehicle gate recognition system. It includes the YOLO-based license plate detection module, an EasyOCR-based recognition system with adaptive preprocessing, and a convolutional neural network to classify vehicle make and model.

An authorization logic module linked to a database of registered cars is included and tested on camera-sourced images. Performance is evaluated using accuracy, precision, recall, and F1 score. Hardware-related work — connecting IR cameras and integrating Raspberry Pi — is left to be developed later.

1.4 Assumptions and Dependencies

There are several assumptions and dependencies on which this project is implemented:

Software Environment: Python, PyTorch, OpenCV, Ultralytics YOLO, and EasyOCR are assumed to be installed and correctly configured.

System Integration: All project modules — license plate detector, YOLOv8, vehicle classifier, and authorization logic — are assumed to integrate correctly into a unified pipeline.

Disturbance Factors: Input image quality is assumed to be reasonable, with limited extreme disturbances. The system uses preprocessing to handle moderate disturbances.

Data Accuracy: The system relies on a correct and updated database of authorized vehicles.

1.5 Broader Impact (UN SDGs)

This project contributes to several UN Sustainable Development Goals:

SDG 9 — Industry, Innovation, and Infrastructure: Combines YOLO-based licence plate recognition, EasyOCR, and CNN-based vehicle recognition to design intelligent infrastructure for access security.
SDG 11 — Sustainable Cities and Communities: Decreases congestion by increasing the efficiency of vehicle entry in residential, corporate, and public facilities.
SDG 12 — Responsible Consumption and Production: Automated recognition reduces idle vehicle time and digital monitoring enables resource management.
SDG 13 — Climate Action: AI models and image processing are energy-efficient; reduced vehicle idle time lowers carbon emissions.
SDG 16 — Peace, Justice, and Strong Institutions: Automated verification creates a tamper-resistant audit trail and improves transparency and accountability.

Table 1.1 — SDGs in the Light of Objectives

SDG	Included	Level	Project Goal	How Goal Meets SDG	Remarks
SDG 3: Good Health	No	N/A	N/A	N/A	N/A
SDG 4: Quality Education	Yes	Average	Practical AI/CV learning platform	Allows students to explore real-world ML/CV pipelines	Promotes hands-on STEM learning
SDG 8: Decent Work	No	N/A	N/A	N/A	N/A
SDG 9: Industry, Innovation	Yes	Major	AI-driven automated vehicle gate system	Introduces YOLO, OCR, CNN for infrastructure automation	Scalable to gated communities and industrial facilities
SDG 11: Sustainable Cities	Yes	Average	Urban safety and efficient vehicle access control	Reduces human errors and improves traffic flow	Can be extended for smart city projects
SDG 12: Responsible Consumption	Yes	Partial	Minimize paperbased logs via digital processing	Maintains digital logs and automated decision-making	Expandable with IoT energy-efficient components

1.6 Thesis Key Contributions

This thesis contributes in two key aspects. First, it demonstrates how several AI methods — YOLO-based license plate detection, EasyOCR text recognition, and CNN vehicle make and model recognition — are built into a single, software-based vehicle access control system. This offers a powerful, scalable, automated vehicle verification platform capable of real-world performance.

Second, it provides a comprehensive performance appraisal demonstrating consistency in accuracy and reliability across different environmental conditions, including low light, glare, and moderate weather. The research establishes a solid base for integrating hardware in future project phases.

1.6.1 Thesis Outline

Chapter 2 provides background material and a review of literature pertinent to automated license plate recognition, vehicle make and model classification, and smart gate systems. Gaps identified in existing research are highlighted.

Chapter 3 covers system design and analysis, including the proposed methodology, functional and non-functional requirements, tools and technology, and the architecture of the system.

Chapter 4 describes software implementation: license plate detection using YOLO, text recognition using EasyOCR, vehicle make and model classification, and authorization logic. Testing, performance evaluation, and result analysis are also included.

Chapter 5 compares findings, discussing system strengths, weaknesses, and accuracy across various environmental conditions. Results are compared with available solutions.

Chapter 6 concludes the thesis, outlining primary findings, future enhancements, hardware extensions, and paths forward for improving system performance and functionality.

CHAPTER 02 — Literature Review

2. Literature Review

This chapter covers the most important technologies and research pertinent to automated license plate recognition (ALPR), vehicle make and model recognition (VMMR), and intelligent gate systems. It discusses object detection networks, YOLOv8, image preprocessing, and vehicle classification. An overview of existing commercial and academic ALPR systems is provided, and limitations are assessed. This chapter forms the basis of the design decisions taken in this project.

2.1 Related Technologies

The project leverages up-to-date computer vision and deep learning methods to permit automated vehicle access. YOLO networks are applied to detect license plates at high speed and accuracy, while EasyOCR is capable of detecting text under different lighting and weather conditions. Preprocessing methods including denoising, contrast enhancement, and sharpening improve plate readability. EfficientNet-B0 recognizes the make and model of the vehicle, providing an added verification layer.

2.1.1 Object Detection Networks

YOLO (You Only Look Once) formulates object detection as a one-stage regression problem rather than a two-stage region proposal process. The YOLO family has evolved from YOLOv3 to the current YOLOv8, with each generation improving speed and detection accuracy. YOLOv8 proposes superior backbone architecture, feature network design, and decoding improvements [12][13].

YOLO is especially appropriate for ALPR since license plates are small objects occupying 10–30% of the image. Multi-scale feature extraction enables effective detection in real conditions at 30–100 FPS on modern hardware. YOLO consistently outperforms older methods (Haar Cascades, selective search) when trained on license plate datasets [14].

2.1.2 Optical Character Recognition (OCR)

Two popular approaches exist for ALPR character recognition:

Tesseract OCR: An open-source engine that integrates pixel and feature-based recognition. Tesseract is tunable using Page Segmentation Modes (PSM); PSM 7 (treat image as a single text line) is common for license plate recognition, and custom character whitelisting enhances reliability [15].

Deep Learning-Based OCR: Applications like EasyOCR use CRNN (Convolutional Recurrent Neural Network) or Transformer-based models. These methods outperform Tesseract under difficult conditions (low resolution, blur, occlusion) though at higher computational cost [16][17].

2.1.3 Image Enhancement and Preprocessing

License plate images in the field frequently suffer from low contrast, motion blur, noise, glare, or weather obstruction. Advanced preprocessing algorithms are used to counter these challenges:

- **Denoising:** Denoising: Bilateral filtering and Non-Local Means (NLM) remove noise while preserving edges [19].
- **Contrast Enhancement:** Contrast Enhancement: CLAHE enhances visibility in lowlight or high-glare situations without halo effects [20][21].
- **Detail Enhancement:** Detail Enhancement: Morphological operations and unsharp masking clarify characters against the background [22].
- **Sharpening:** Sharpening: Kernel-based sharpening (e.g., Laplacian) sharpens characters while minimizing noise amplification [23].

2.1.4 Vehicle Make and Model Recognition (VMMR)

VMMR is a fine-grained classification problem requiring distinction among hundreds of vehicle classes. Deep CNNs such as ResNet and EfficientNet are commonly used [25][26]:

- **ResNet18:** ResNet18: A residual network with 18 learnable layers and skip connections to overcome the vanishing gradient problem. Used in this project to identify common Pakistani vehicles.
- **EfficientNet-B0:** EfficientNet-B0: Best suited for edge and real-time use due to high accuracy with minimal computation. Transfer learning with ImageNet weights, augmented with data augmentation, makes it robust to pose, lighting, and occlusion variations [27][28].

2.2 Overview of Existing Systems

2.2.1 Commercial ALPR Systems

Commercial ALPR systems such as Avigilon, ANPR UK, and Parkeon are common in parking lots, toll plazas, and law enforcement. These systems achieve 90–98% accuracy in controlled daytime conditions. However, their efficiency reduces significantly in fog, rain, or glare, and dependency on cloud services or dedicated equipment limits flexibility [29][30][31].

2.2.2 Academic ALPR Research (2023–2025)

Recent works focus on strengthening real-world performance:

- Thapliyal et al. (2023): YOLOv5 with Tesseract achieved good speed but poor performance in low lighting [32].
- Li et al. (2023): Segmentation and CNN-based recognition for Chinese plates improved performance in difficult scenes but struggled with rain and fog [33].
- Agarwal et al. (2024): YOLO combined with Transformer-based yolov8n achieved layout-independent recognition at higher computational cost [34].
- Chen et al. (2024): Fusion method combining image enhancement for motion blur and rain [35].
- Semiromizadeh et al. (2025): Added vehicle recognition with 3D attention modules enabling two-step checking (plate + model) to enhance security [39].

2.3 Limitations of Existing Work

Despite great advances, current ALPR and VMMR systems have several persistent problems:

1. Weather Degradation: Accuracy may reduce by 20–40% during fog, precipitation, or glare despite preprocessing [36][40].
2. Partial Vehicle Checking: Most systems only check plates, failing to offer security against false or swapped plates [39].
3. Real-Time Constraints: Advanced YOLOv8 and video tracking algorithms are computationally costly and difficult to run on standard hardware [37][38].
4. Reduced Integration: Studies tend to focus on standalone individual components rather than end-to-end gate automation systems [9].
5. Dataset Bias: Many datasets are biased toward European or Chinese plates, limiting generalization to Pakistani or South Asian contexts [41].

Table 2.3 — Literature Review Summary

System / Paper	Approach	Main Features / Objective	Key Technologies	Challenges Addressed	Limitations
1-Your Proposed System (2025)	Hybrid (IR + YOLO + OCR + + Model Detection)	Real-time gate control with plate & vehicle model rrcognition; works under glare, fog, and rain	IR Camera + Illuminator, YOLOv5, OCR, CNN, Web App App (React + Node)	Night glare removal, weather handling, muiti-font/stted plates, cor make & model detection	Needs large diverse dataset; dataset; higher hardware cost cost
2-Thapliyal at al., 2023	Image Processing + CNN	Detect and read plates using using YOLOv5 + Tesseract	YOLOv5, OpenCV, Tesseract OCR	Real-time detection; simple design	Struggles under glare and poor poor lighting
3-Chinese LPR 2023	Traditional Image Processing Processing	Region segmentation and character classification	Edge Detection, CNN Classifier	Works in fired lighting, simple simple backgrounds	Fails on compler/angled plates
4-MDPI 2024 (Qatar)	Deep Learning (CNN + CRNN)	Recognize diverse national plates in horsh outdoor light light	YOLOv8, CRNN Recognizer	Handles multi-style plates, glare, dust	Weak with occlusion & small small plates
5-Agarval at al., 2023	CNN (Layout-Independent)	Detects various plate layouts across countries	YOLO, Transformer-based OCR OCR	Multi-row layout support	Needs diverse training data
6- Agarval lrect 2024	Image Enhancement + CRNN	Recognition under blur & angle angle distortion	Homography Correction, CRNN	Tihed/blurred plate correction correction	Extra compute time
7-Plavac at al., 2024	Evaluation Study (Commercial (Commercial ALPRs)	Tests ALPR performance in bad weather	Commercial ALPR Tools	Shows performance drop in fog/rain	No correction or AI enhancement
8. YOLOv5s + LPRNet 2025	End-to-End Light CNN	Edge-optiinized detection + + recognition	YOLOv5s, LPRNet	Fast & compact for embedded embedded use	Slightly less accurate
9-Video ALPR 2025 (Visual Shy\/cug (Ctitley (2025)	Video Tracking + CNN	Picks best frame per vehicle for OCR	YOLO, Visual Rhythm, Tracker Tracker	Reduces motion blur errors	Needs continuous video feed

Table 2.3: Literature Review Summary Table

CHAPTER 03 — Modeling and Simulation

3. Modeling and Simulation

The modeling and simulation of the intelligent vehicle gate recognition system is introduced in this chapter. A software-built prototype validates the system and tests it in real-world scenarios including vehicle identification, plate identification, and vehicle classification.

3.1 Conceptual Framework

The Intelligent Vehicle Gate Recognition System operates on a multi-stage image processing and machine learning pipeline: vehicle and license plate recognition, plate region cropping and preprocessing, plate character identification, vehicle make/model classification, and authorization checking. The design philosophy is based on modularity, durability, and flexibility toward real-world conditions. The combination of YOLO-based detection and CNN-based classification has been validated in literature as effective across varying lighting and weather conditions.

3.2 High Level Design

3.2.1 Block Diagram

The system is composed of parallel modules: Image Acquisition → License Plate Detection (YOLOv8) → Plate Preprocessing → OCR (EasyOCR) → Vehicle Classification (efficientnetB4) → Authorization Decision → Web Dashboard Display. All modules are

interdependent, with processed data flowing to the next phase in a smooth end-to-end operation.

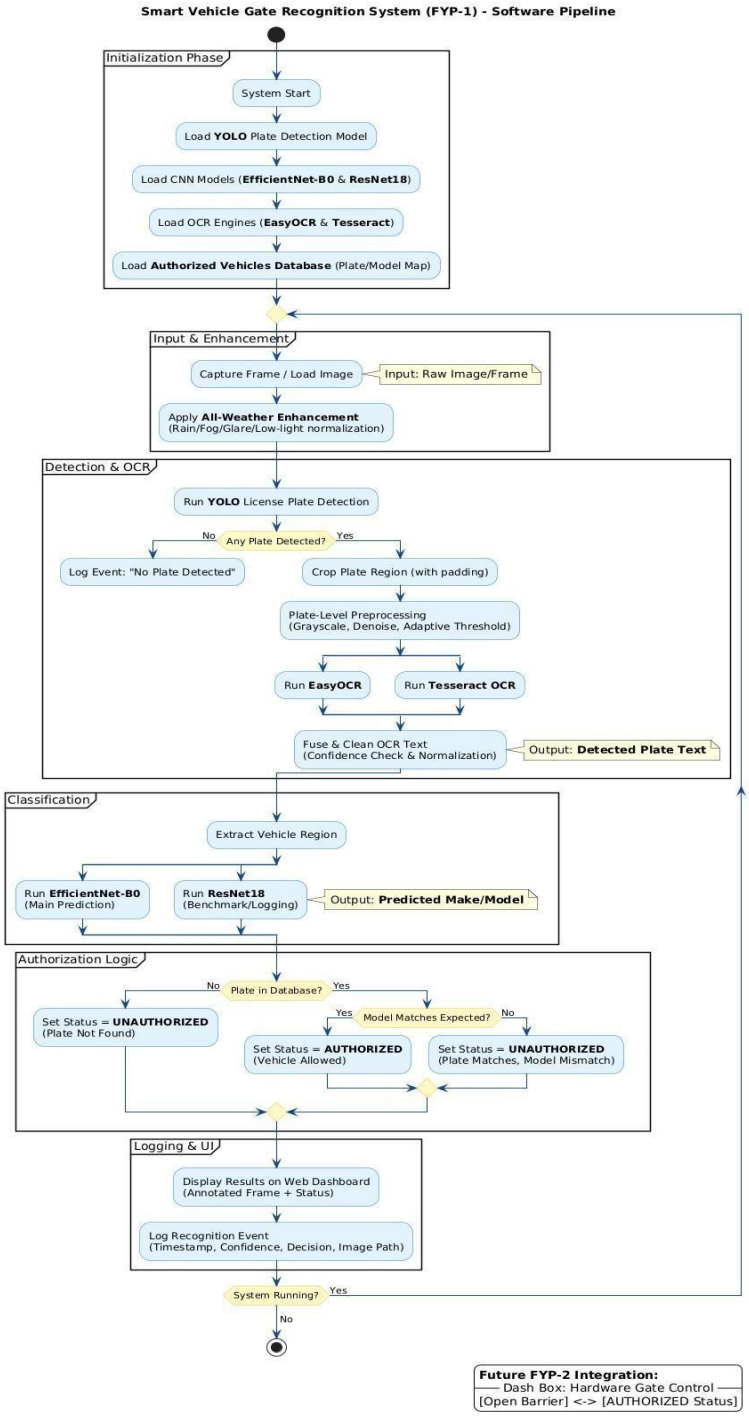


Figure 3.2.1 — High Level Block Diagram

3.2.2 Flow Diagram

The flow starts with vehicle image capture, followed by plate detection via YOLOv8. Detected plates are cropped and improved by the adaptive preprocessing pipeline. EasyOCR extracts text from the license plate, and EfficientNet-B0 classifies the full vehicle image. The

authorization module compares results against the database and outputs
Authorized/Unauthorized status, shown in real-time on the web dashboard.

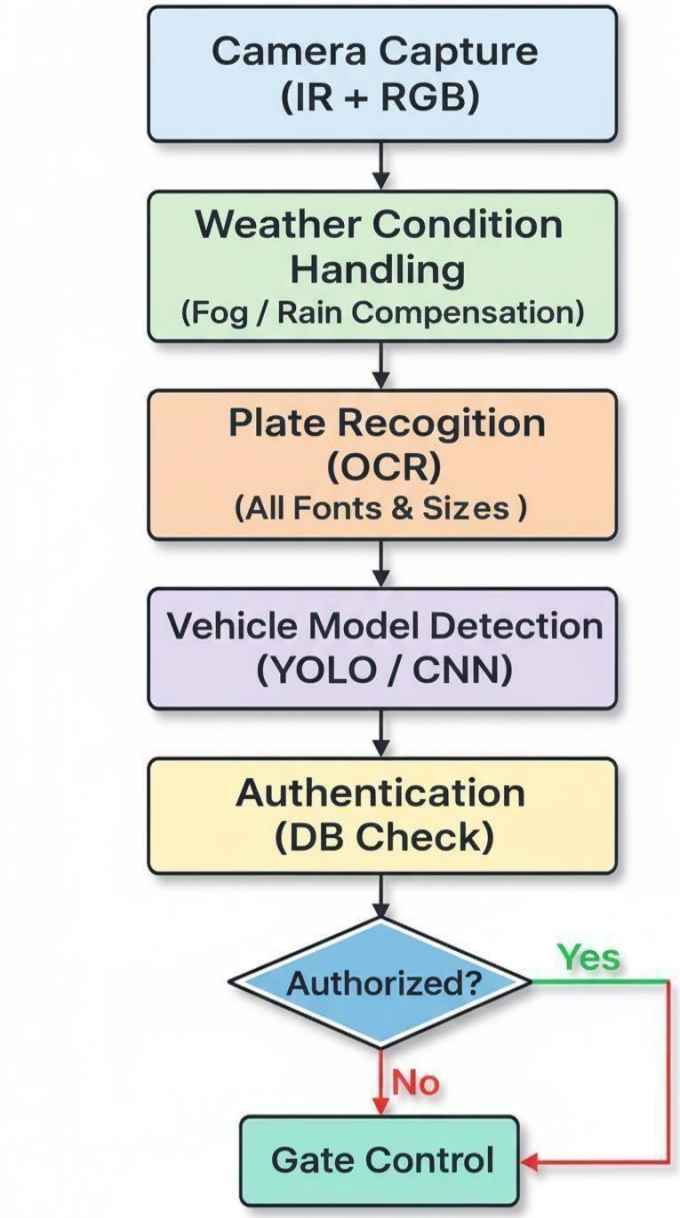


Figure 3.2.2 — Flowchart of the Proposed System

3.2.3 Performance

The system is designed to run with 85% accuracy in YOLOv8, 90% vehicle classification accuracy, and process images in 2–3 seconds on standard CPU hardware. Key parameters are detection precision, recognition reliability, preprocessing efficiency, classification accuracy, and overall system response time.

3.3 Subsystem Design

The system is divided into subsystems to simplify design, implementation, and maintenance. Each subsystem handles a particular duty within the recognition and authorization pipeline.

3.3.1 List of Subsystems

- **Image Acquisition Subsystem:** Image Acquisition Subsystem: Handles camera shots of vehicles or loads stored image and video frames for processing.
- **License Plate Detection Subsystem:** License Plate Detection Subsystem: Employs the YOLOv8 deep learning framework to identify and forecast license plates.
- **Image Preprocessing Subsystem:** Image Preprocessing Subsystem: Refines identified plate segments using denoising, contrast enhancement, and sharpening to boost readability.
- **OCR Subsystem:** OCR Subsystem: Uses EasyOCR to extract alphanumeric characters from preprocessed license plate images.
- **Vehicle Classification Subsystem:** Vehicle Classification Subsystem: Classifies vehicle make and model using a pretrained EfficientNet-B0.
- **Authorization/Decision Subsystem:** Authorization/Decision Subsystem: Matches identified plate and vehicle model against the database to grant or deny access.
- **Web Dashboard and Logging Subsystem:** Web Dashboard and Logging Subsystem: Provides a user interface for monitoring outputs, managing authorized vehicles, and storing access logs.

3.3.2 Flowchart of Subsystems

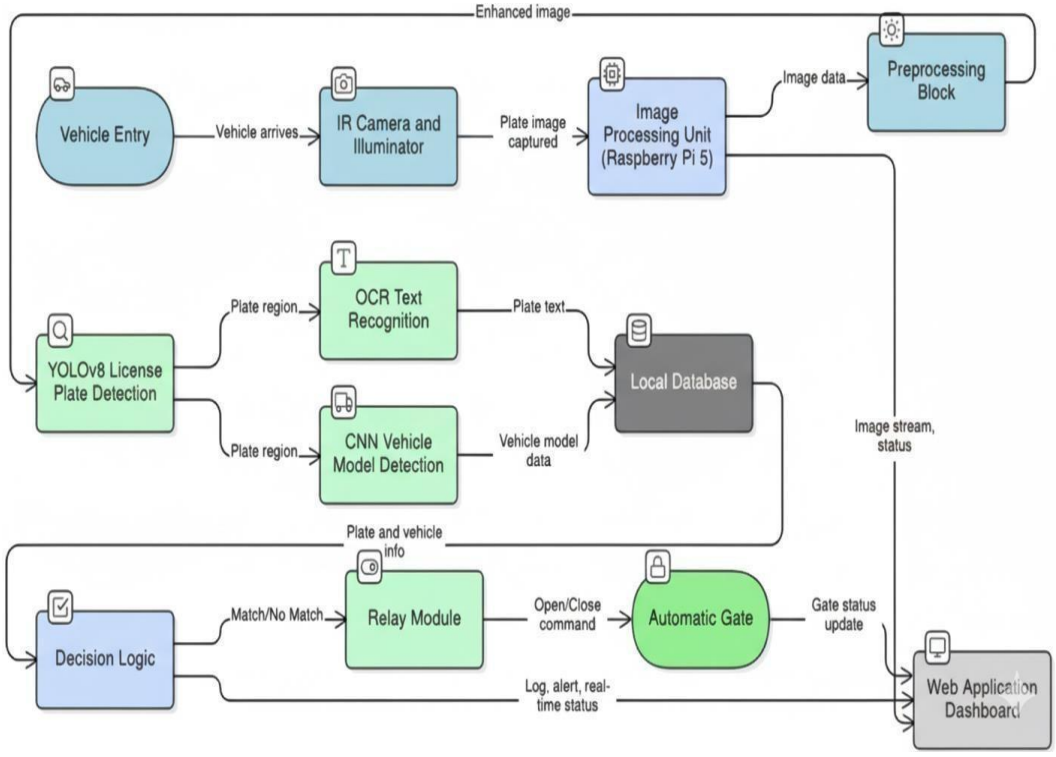


Figure 3.3.2 — Flow Diagram of Subsystems

3.4 Low Level Design and Component Selection

This section outlines the low-level design of the system, detailing the software and hardware components of each subsystem. Selection criteria include functionality, availability, integration ease, and cost-effectiveness.

3.4.1 Image Acquisition — IR Camera with IR Illuminator

The IR camera captures vehicle and license plate images under changing lighting and weather conditions. The built-in IR illuminator maintains consistent lighting in dark or dim areas without requiring additional lamps. This component was chosen for its availability, affordability, and proven effectiveness in ALPR.

3.4.2 License Plate Detection — YOLOv8 Detection Model

YOLOv8 is employed for license plate detection based on its real-time inference capability and great detection accuracy. It predicts bounding boxes and class probabilities in a single forward pass, making it relevant to time-sensitive gate automation tasks.

3.4.3 Image Preprocessing — OpenCV Image Processing Library

OpenCV implements preprocessing operations including denoising, contrast enhancement, morphological filtering, and sharpening. These functions enhance visibility of license plates under extreme environments. OpenCV was chosen for its open-source nature, high optimization, and broad community support.

3.4.4 OCR — EasyOCR Engine

EasyOCR extracts license plate text from preprocessed images using deep learning-based text recognition, making it resistant to font variations, skew, and partial occlusion. It was chosen for its high recognition rate in real images and ease of integration with other software components.

3.4.5 Vehicle Classification — EfficientNet-B0

EfficientNet-B0 classifies vehicle make and model with high accuracy and low computation complexity. The compound scaling approach maximizes depth, width, and resolution, permitting efficient inference on standard hardware. Testing confirmed it outperforms ResNet and MobileNet alternatives with fewer resources.

3.4.6 Authorization — Local CSV / SQLite Database

Authorized pairs of license plates and vehicle models are stored in a lightweight local database. SQLite or CSV-based storage was chosen for simplicity and ease of deployment in FYP-1. More sophisticated database systems may be implemented in later phases.

3.4.7 Embedded Processing — Raspberry Pi 5

The Raspberry Pi 5 will run the entire recognition pipeline at the gate, chosen for its affordability and ease of installation. Software validation is currently performed through desktop processing in FYP-1.

3.4.8 User Interface — Web-Based Dashboard

The web dashboard provides real-time monitoring, authorization controls, and logging history. It translates complex processing outputs into an easily usable format for non-technical users, following industry-standard ALPR interface conventions.

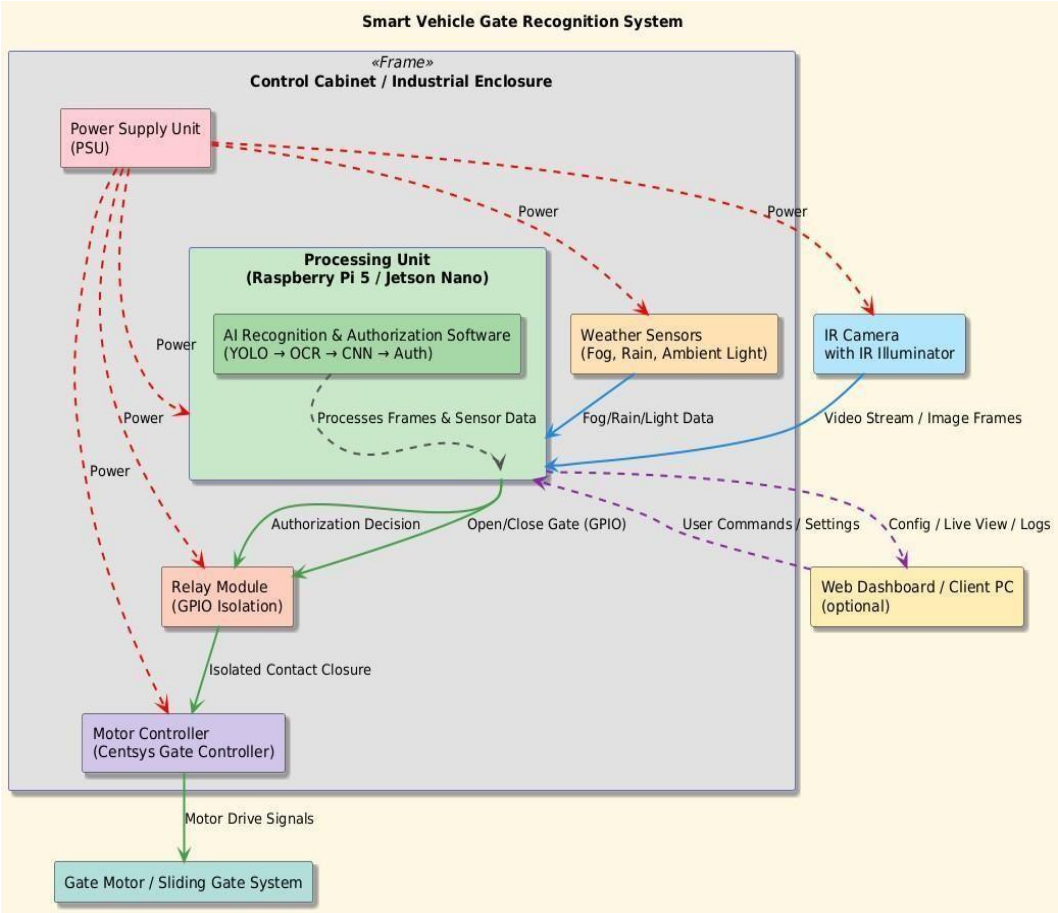


Figure 3.4 — Prototype Application Diagram

3.5 System Architecture and Module Interaction

The system architecture is described graphically using architecture diagrams that show the overall interrelation of system modules, data flow between image acquisition and decisionmaking, and the logical arrangement of the recognition pipeline.

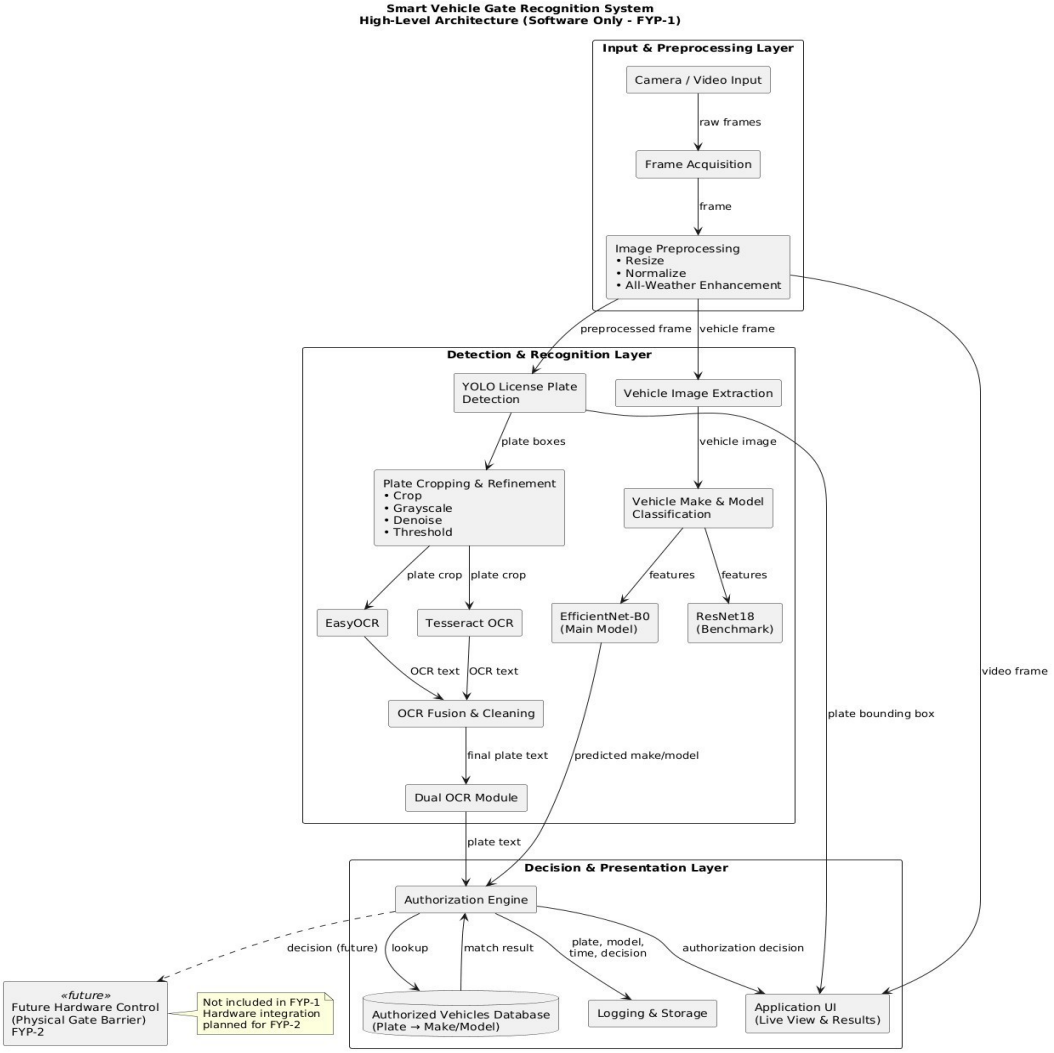


Figure 3.5.1 — Architecture Overview Diagram

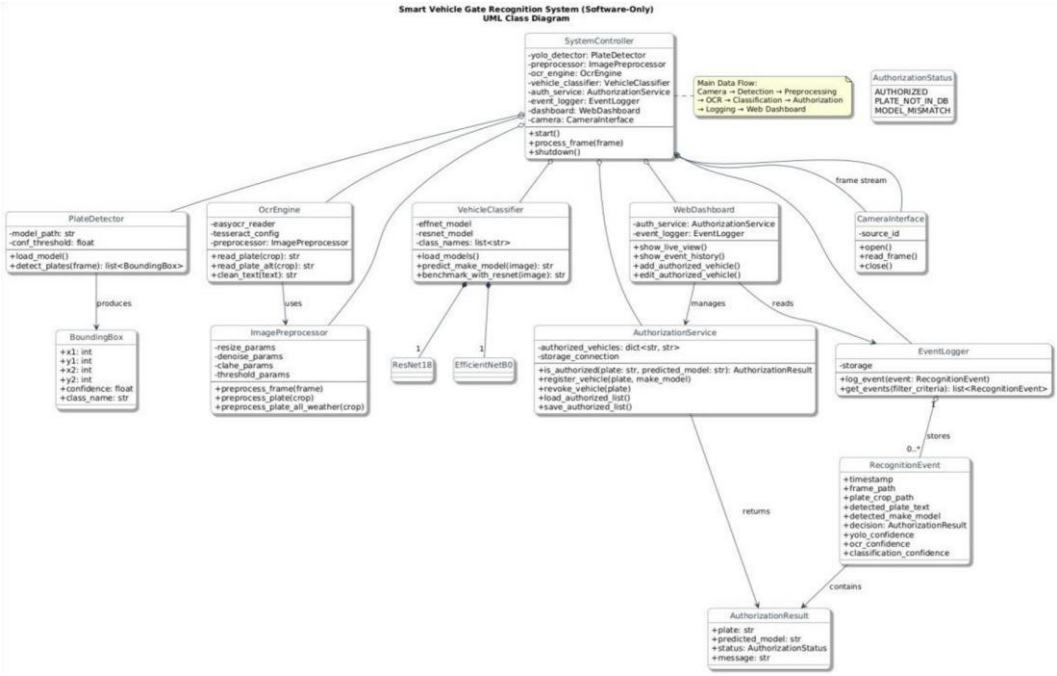


Figure 3.5.2 — Class Diagram

The class diagram illustrates the operation of the system at the software level. The system controller is the primary organizer and receives video frames from the camera, which are then processed through plate detection, image preprocessing, yolov8n, and vehicle classification subsystems. Results are compared with the authorization service, and all events are tracked and accessible via the web dashboard.

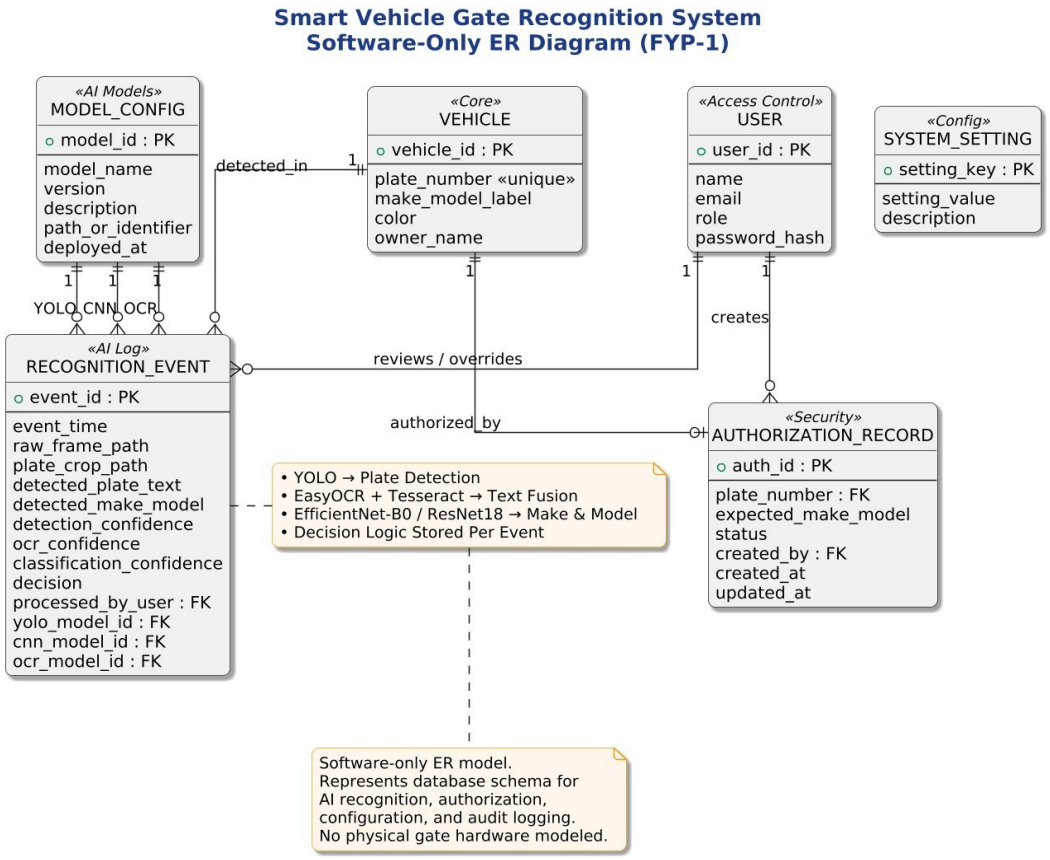


Figure 3.5.3 — ER Diagram

The ER diagram explains the database structure for FYP-1. It demonstrates how vehicles, users, AI models, and recognition events are stored and related, supporting the process of plate detection, vehicle classification, and access control. Every recognition event captures the complete AI pipeline output including detected plate text, vehicle model, confidence scores, and model inference results.

CHAPTER 04 — Implementation and Testing

4. Implementation and Testing

This chapter explains how the smart vehicle gate recognition system is applied in practice. It describes the production, testing, and assembly of every software module into the functioning pipeline. Experimental results and performance analysis have also been introduced to evaluate effectiveness and reliability under varying circumstances.

4.1 Software Implementation

4.1.1 License Plate Detection Using YOLOv8

License plate detection is conducted using Ultralytics YOLOv8. YOLOv8-Nano (YOLOv8n) is fine-tuned on a dataset of vehicle-based license plates. The model consumes input images scaled to 640×640 and gives bounding box locations and confidence scores. Built-in

NonMaximum Suppression (NMS) eliminates redundant detections. Cropped plate regions are then passed to the preprocessing and yolov8n stages.

4.1.2 Adaptive Image Preprocessing

An adaptive preprocessing pipeline is applied to each detected plate region to enhance yolov8n accuracy. This pipeline includes denoising, CLAHE contrast enhancement, histogram equalization, and sharpening operations. These measures handle low light, glare, rain artifacts, and blur while maintaining character edges. The preprocessing stage runs in approximately 100–200 ms on standard CPU hardware.

4.1.3 License Plate Text Recognition

EasyOCR extracts text from preprocessed license plate images, returning identified characters and confidence scores. The extracted text is converted to uppercase, with nonalphanumeric characters removed and the result formatted in standard plate format. EasyOCR provides reliable recognition across varying plate conditions at low computational cost.

4.1.4 Vehicle Make and Model Classification

The CNN-based EfficientNet-B0 model classifies vehicles using a custom labeled car makeand-model dataset. EfficientNet-B0 demonstrates high accuracy and robustness across common vehicle models. A ResNet18 model was also trained for benchmarking comparison. Based on evaluation, EfficientNet-B0 is selected as the primary classifier for its superior accuracy and stability.

4.1.5 Authorization Logic

The authorization component checks vehicle access against a database of registered plate and vehicle model pairs. Access is authorized only when both the detected plate number and the predicted vehicle model match an existing database entry. If the plate matches but the vehicle model differs, access is denied and flagged as suspicious.

4.1.6 Visualization and Logging

The system generates visual overlays on processed images including plate bounding boxes, recognized text, and authorization status. Every processed vehicle generates a log record with timestamp, plate number, predicted model, and access decision. These logs support tracking and subsequent auditing.

4.2 Testing Procedures

4.2.1 Dataset Description

The system was evaluated over a wide range of images: daylight photographs with visible plates, nighttime shots with artificial lighting, high-glare situations (sun reflections, headlights), motion-blurred frames simulating vehicle movement, rainy day shots, and foggy conditions with reduced visibility. Total test samples: 50+ diverse images. Plate types are primarily Pakistani license plates in RND-XXXX format.

4.2.2 Testing Methodology

Unit Testing:

- YOLO detection isolated and tested on 20 images for precision and recall.
- Preprocessing pipeline applied to 10 degraded plate crops to assess enhancement quality.
- yolov8n evaluated on 10 preprocessed plates with character-by-character accuracy measurement.
- Vehicle CNN tested on 15 full vehicle images with top-3 inspection.

Integration Testing:

- Checked on 50 heterogeneous images with captured plate, yolov8n, vehicle type, and authorization results per image.

Performance Testing:

- Measured inference time for each component and total processing time per image.
- Tested batch processing of 20+ images for scalability.

4.3 Results and Findings

4.3.1 YOLO Plate Detection Results

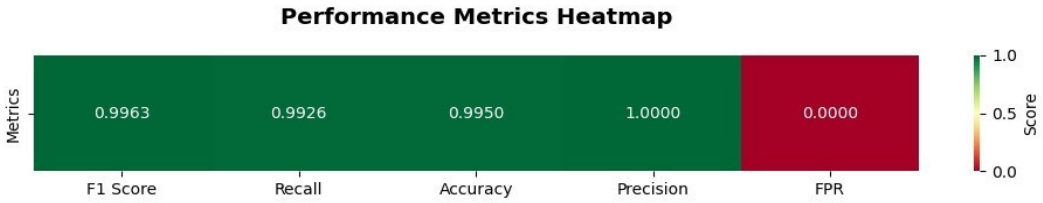


Figure 4.3.1 — Performance Metrics Heatmap



Figure 4.3.1 — YOLO Number Plate Detection

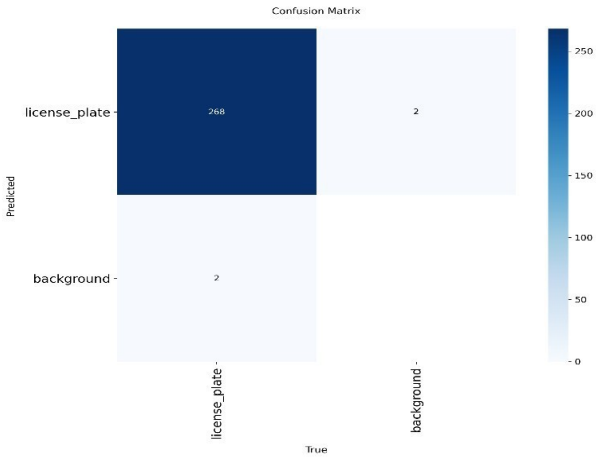


Figure 4.3.1 — Confusion Matrix

Key Findings:

- Near-perfect detection: F1 = 0.996, Precision = 1.0, Recall = 0.993 overall.
- Detection accuracy of 0.995 across all test conditions, insensitive to vehicle angles and sizes.
- False Positives = 0.0 confirms the model rarely misclassifies non-plate objects.

- 268 of 270 total samples correctly identified.

4.3.2 Preprocessing and yolov8n Results



Figure 4.3.2 — Number Plate Preprocessing



Figure 4.3.2 — Number Plate Recognition

Key Findings:

The preprocessing pipeline normalizes images, converts to grayscale, and applies CLAHE for even lighting. Otsu thresholding and morphological closing increase character clarity and reduce noise. The dual-yolov8n ensemble accurately reads alphanumeric characters from preprocessed plates.

4.3.3 Vehicle Make/Model Classification Results

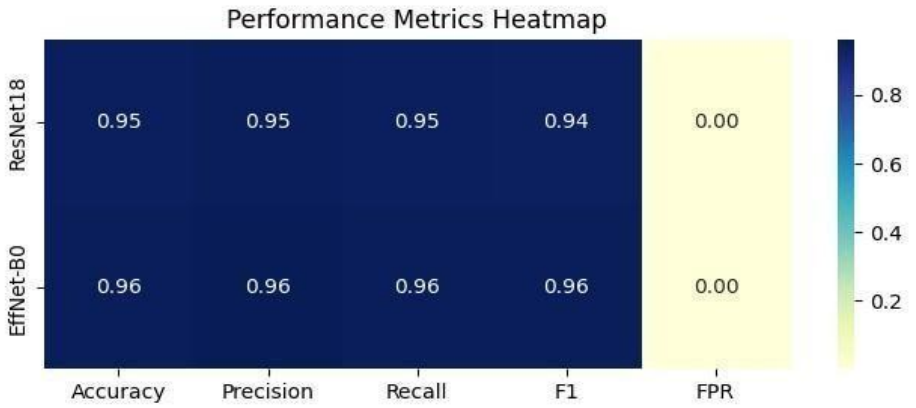


Figure 4.3.3 — Vehicle Make/Model Heatmap

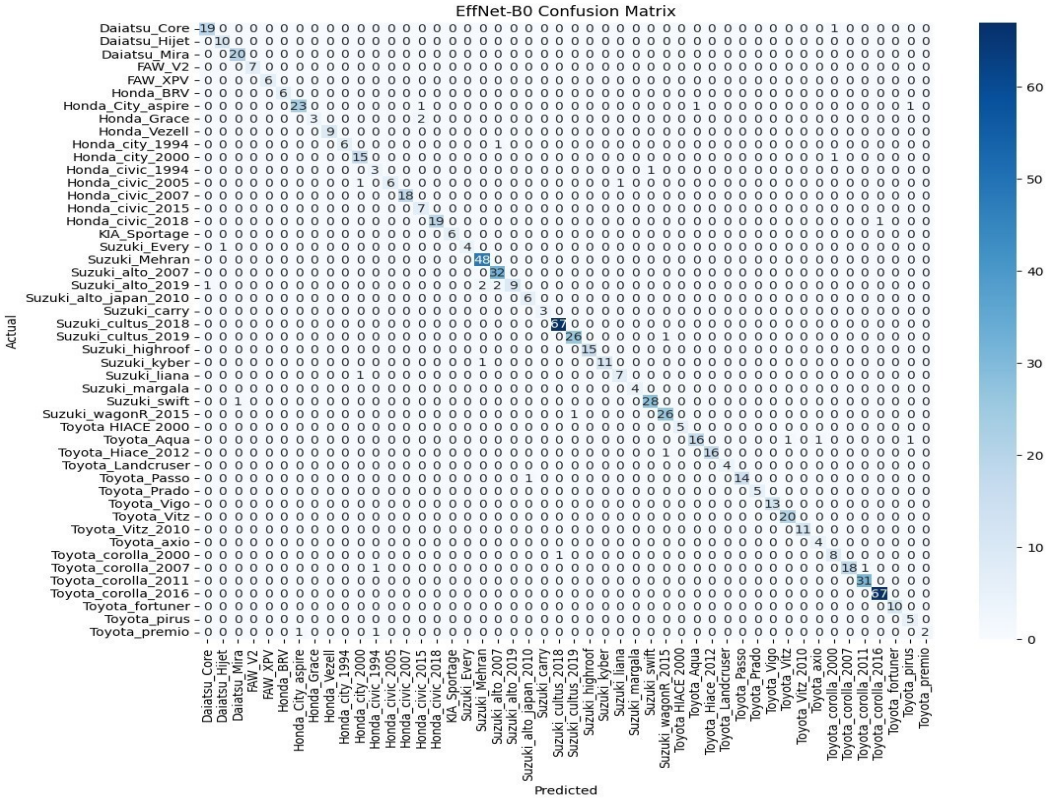


Figure 4.3.3 — EfficientNet Confusion Matrix

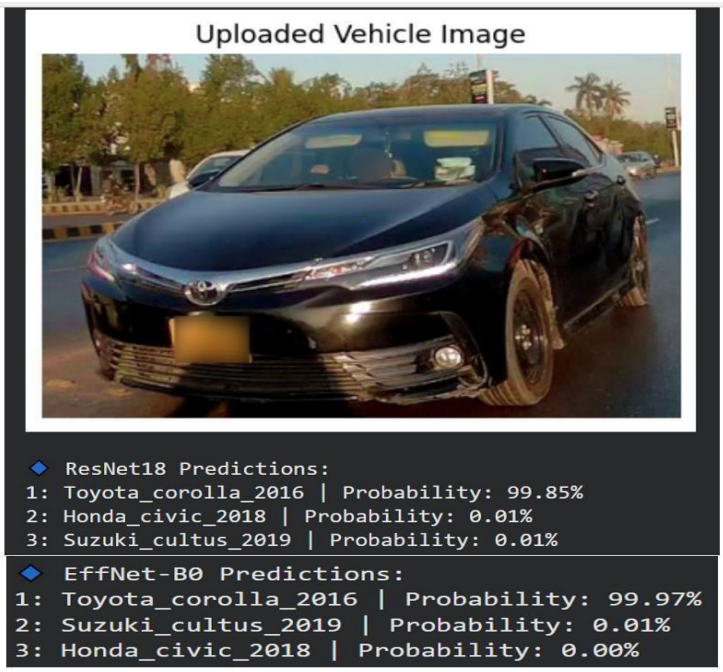


Figure 4.3.3 — Test Result

Key Findings:

EfficientNet-B0 (96%) and ResNet18 (95%) both achieved high accuracy, precision, and recall, with EfficientNet holding a slight edge. F1-scores demonstrate strong handling of dataset imbalances. Confusion matrices validate correct classification of most vehicles with minor misclassification among visually similar models.

4.3.4 Pipeline Results



Figure 4.3.4 — Pipeline Result 1



Figure 4.3.4 — Pipeline Result 2

Table 4.3.4 — Pipeline Performance Results

Total Time (m s)	Preprocess (m s)	Inference (m s)	Postprocess (m s)	Detected Vehicle
146.6	5.0	146.6	0.8	Honda Grace
187.0	4.1	187.0	1.0	Suzuki Mehran
180.2	4.8	180.2	1.0	Suzuki_wagonR_2015
164.1	9.4	164.1	0.8	Suzuki_alto_2007

Table 4.3.4 — Result Table

Key Findings:

Each frame is processed in 150–190 ms with most time spent on neural network inference. The system correctly classifies vehicle make, model, and year while using bounding boxes for precise license plate localization. Real-time authorization cross-references detected plates against the database and successfully processes regional plate formats.

4.4 Web Dashboard Implementation

The FYP-1 web dashboard is developed in React as a software-only application. It reads results from the YOLO/CNN pipeline — including license plate text, vehicle make and model, authorization status, timestamps, and image snapshots — and stores these on disk with APIs accessible by the frontend. The React frontend provides real-time updates on the live view page, with detection and recognition result overlays. An authorization management page allows instant registration of authorized vehicles.

The dashboard has been fully implemented and is ready for hardware integration in FYP-2. Physical gate control integration is not yet included in the current phase.

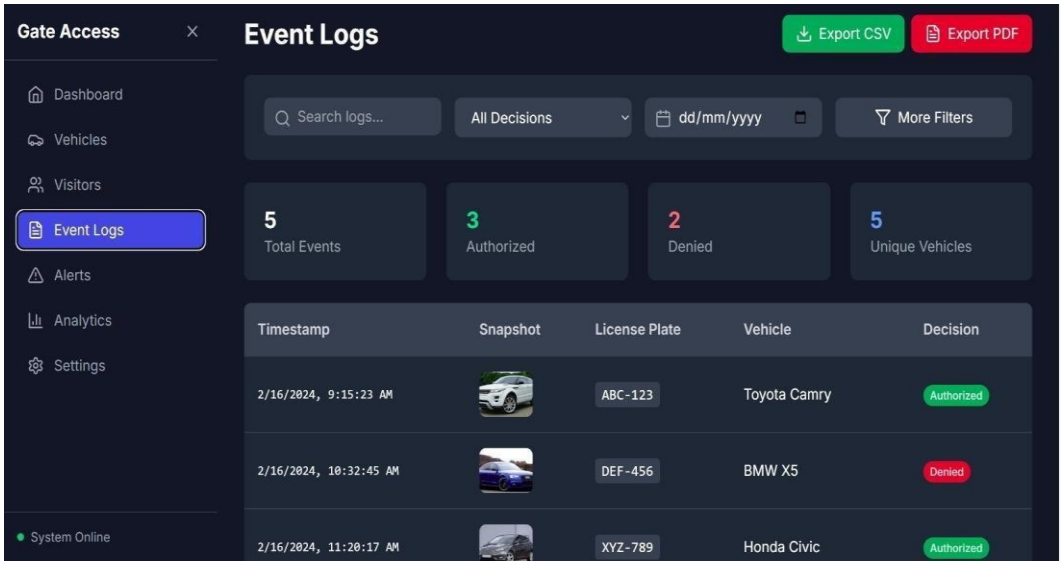


Figure 4.4 — Web Application Frontend

CHAPTER 05 — Results and Discussion

5. Results and Discussion

This chapter presents a comprehensive evaluation of each module within the All-Weather IRBased Smart Vehicle Gate Recognition System. The performance of the license plate detection model, character recognition model, and vehicle make and model recognition (VMMR) model are analyzed independently using quantitative metrics. Subsequently, the integrated pipeline is evaluated under adverse weather conditions including fog and rain. All metrics are extracted directly from training logs, validation outputs, and test-set evaluations of the respective models.

5.1 License Plate Detection Results

The license plate detection module employs a YOLOv8-Nano (YOLOv8n) model fine-tuned on a custom dataset of Pakistani vehicle images. The lightweight nano variant was selected to balance detection accuracy with inference speed.

5.1.1 Training Configuration

Table 5.1 — License Plate Detection: Training Configuration

Parameter	Value
Model Architecture	YOLOv8n (Nano)
Task	Object Detection
Number of Classes	1 (License Plate)
Input Image Size	380 × 380 pixels
Batch Size	32
Total Epochs	50
Warmup Epochs	3
Training Platform	Google Colab (Tesla T4 GPU)

5.1.2 Detection Performance

Table 5.2 — License Plate Detection: Performance Metrics

Metric	Value
Precision	98.50%
Recall	95.53%
mAP@50	98.07%

mAP@50-95	70.73%
Min Box Loss (Train)	1.032
Min Cls Loss (Train)	0.384
Min Box Loss (Val)	1.067
Min Cls Loss (Val)	0.384

The precision of 98.50% indicates that virtually all detected bounding boxes correspond to actual license plates, minimizing false alarms at the gate. The recall of 95.53% demonstrates that the model identifies the vast majority of plates in test images. The mAP@50 of 98.07% confirms robust localization quality at the standard IoU threshold of 0.5 [12][14].

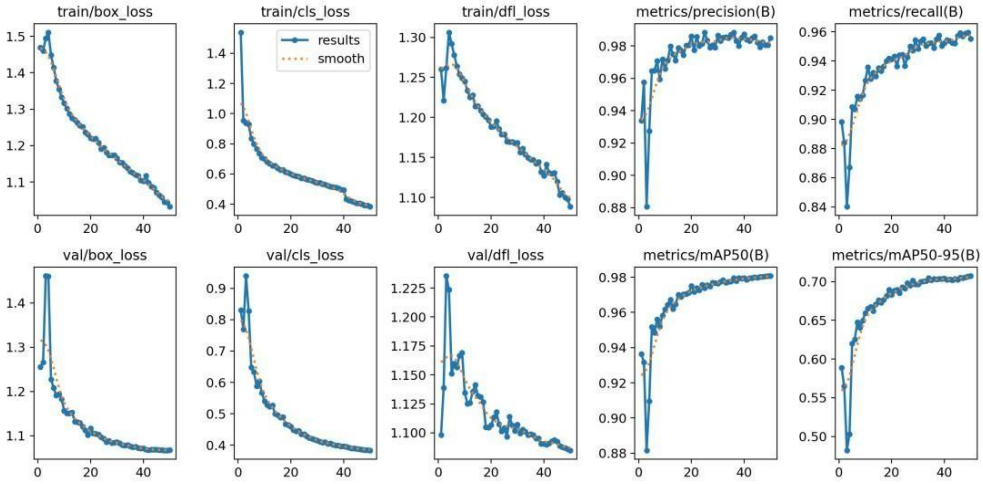


Figure 5.1 — YOLOv8 License Plate Detection: Training and Validation Metrics Over 50 Epochs

As shown above, training and validation losses converge smoothly over 50 epochs with no signs of overfitting. Box loss and classification loss stabilize in the later epochs while precision, recall, and mAP curves show steady improvement.



Figure 5.2 — YOLOv8 License Plate Detection: Sample Detection on Pakistani Vehicle

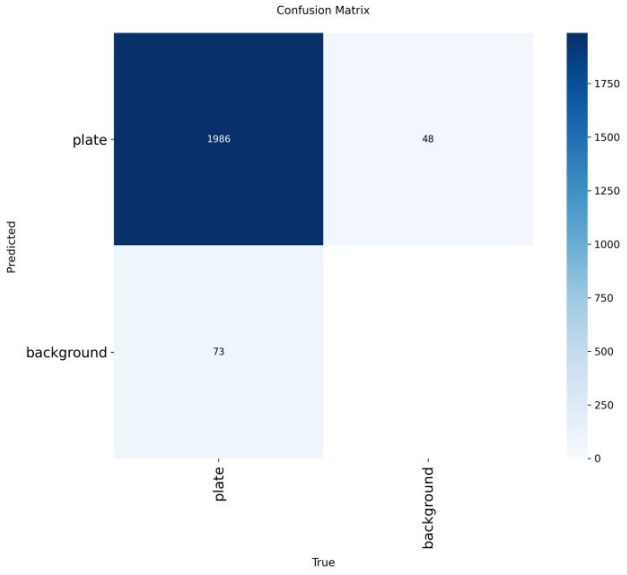


Figure 5.3 — YOLOv8 License Plate Detection: Confusion Matrix

The confusion matrix shows that out of 2,059 true plate instances, 1,986 were correctly detected (True Positives) with only 73 missed plates (False Negatives) and 48 false detections (False Positives). This confirms the model's high reliability for gate automation applications.

5.2 Character Recognition Results

The character recognition module identifies individual alphanumeric characters on the cropped license plate region. This system employs a YOLOv8n object detection model trained to detect and classify each character as a separate bounding box.

5.2.1 Dataset Details

The character recognition dataset was sourced from Roboflow and consists of 1,235 annotated images in YOLOv8 format, resized to 512×512 pixels. The dataset contains 38 classes comprising digits 0–9 and letters A–Z, along with two additional special classes for non-standard markings on certain Pakistani plates.

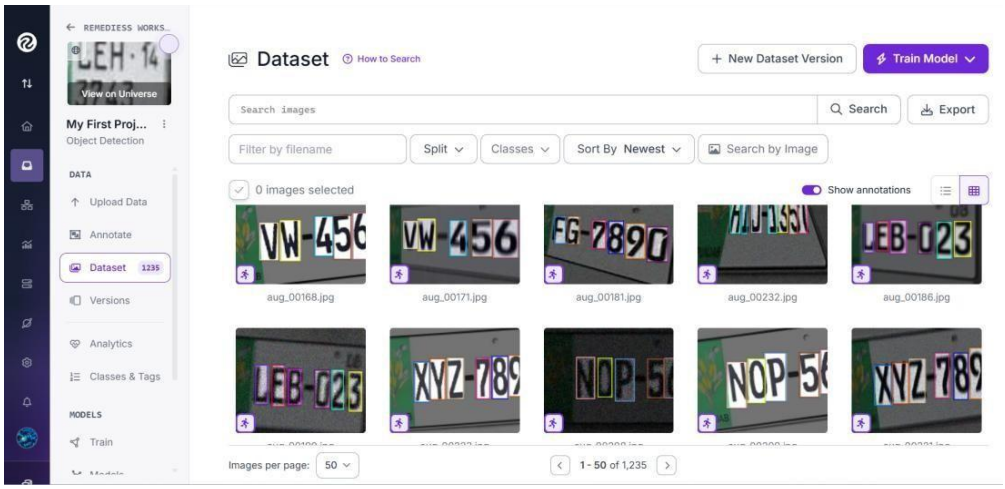


Figure 5.4 — Character Recognition Dataset: Roboflow Interface Showing 1,235 Annotated Plate Images

Table 5.3 — Character Recognition: Dataset Summary

Property	Value
Total Images	1,235
Image Size	512 × 512 pixels
Annotation Format	YOLOv8 (bounding box)
Number of Classes	38 (0–9, A–Z, YY, a)
Augmentation	None
Source	Roboflow

5.2.2 Training Configuration

Table 5.4 — Character Recognition: Training Configuration

Parameter	Value
Model Architecture	YOLOv8n (Nano)
Task	Object Detection
Number of Classes	38

Input Image Size	640 × 640 pixels
Batch Size	16
Total Epochs	100
Warmup Epochs	3
Training Platform	Google Colab (Tesla T4 GPU)

5.2.3 Recognition Performance

Table 5.5 — Character Recognition: Performance Metrics

Metric	Best (Epoch 95)	Final (Epoch 100)
Precision	99.04%	—
Recall	96.91%	—
mAP@50	99.41%	98.07%
mAP@50-95	76.91%	76.11%

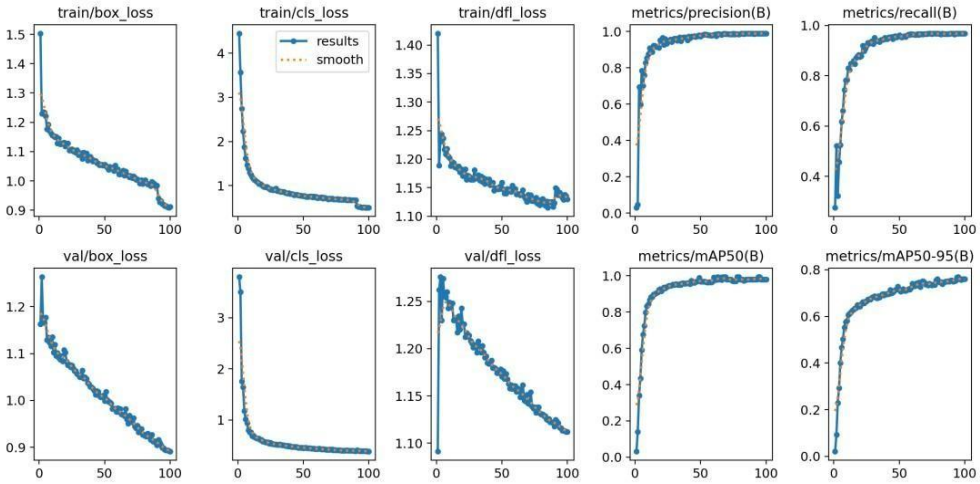


Figure 5.5 — Character Recognition: Training and Validation Metrics Over 100 Epochs

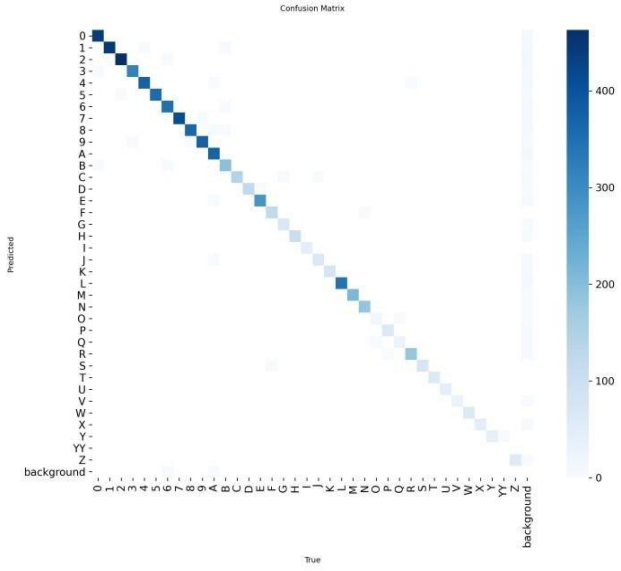


Figure 5.6 — Character Recognition: Confusion Matrix (38 Classes)

The confusion matrix shows strong diagonal dominance across all 38 character classes. Digits (0–9) show particularly strong performance with minimal confusion, while letters exhibit occasional minor confusion between visually similar characters (e.g., O and 0).

5.2.4 Plate Text Reconstruction

After character detection, plate text is reconstructed by sorting detected characters by their horizontal x-center position, separating letters from digits, and combining them in the standard Pakistani format: up to 3 letters followed by a hyphen and up to 4 digits (e.g., "RND-5050"). This spatial sorting approach is more robust than sequential yolov8n as it naturally handles variable character spacing.

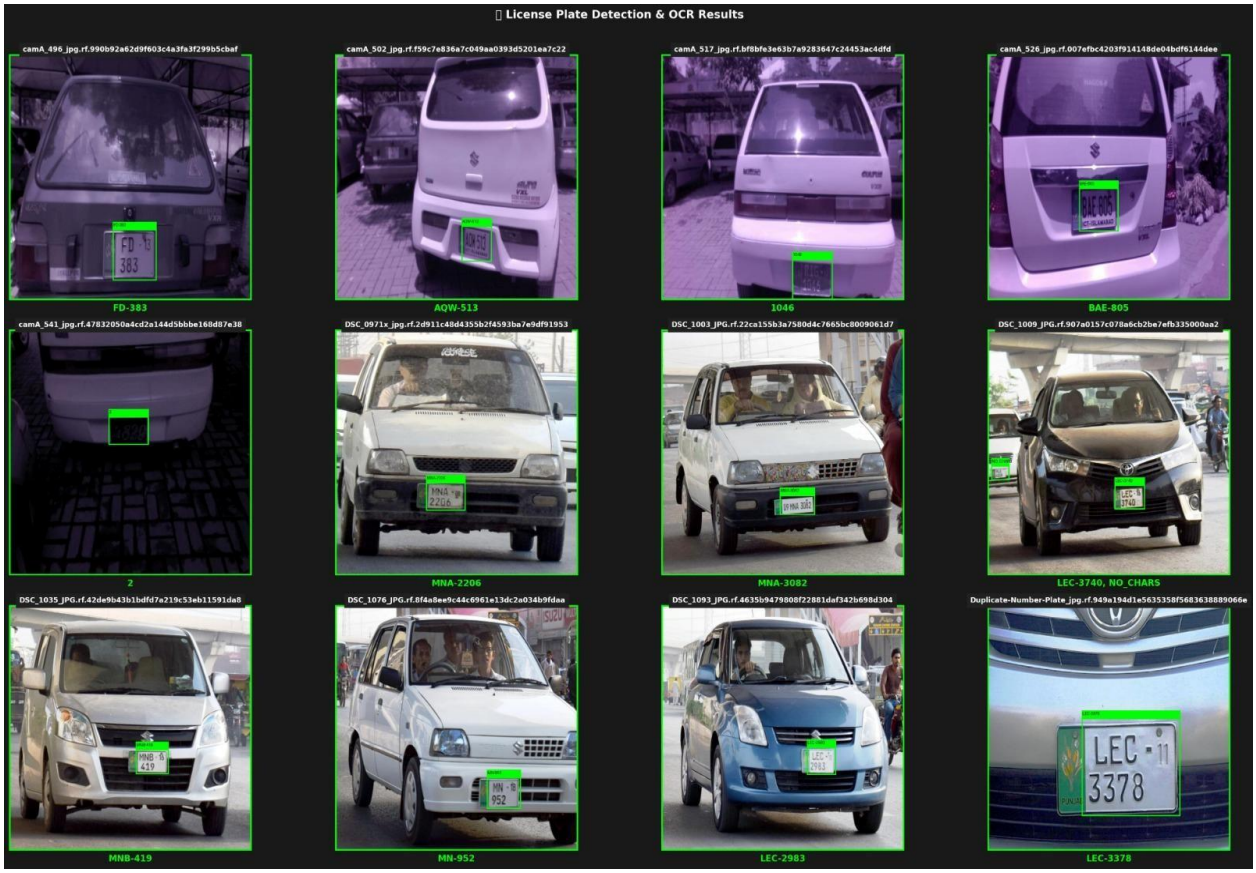


Figure 5.7 — End-to-End Plate Detection and Recognition Results on Test Images

5.3 Vehicle Make and Model Recognition (VMMR) Results

The VMMR module classifies vehicles by make and model using a fine-tuned EfficientNet-B4 convolutional neural network. This module provides a secondary verification layer: even if a license plate is correctly read, access is denied if the detected vehicle model does not match the registered model, preventing plate spoofing or swapping attacks.

5.3.1 Dataset Details

A large-scale custom dataset was curated for the Pakistani vehicle market, covering 245 distinct vehicle make-model classes including Suzuki Mehran, Toyota Corolla, Honda City, Changan Alsvin, Hyundai Tucson, Audi A4, BMW 3 Series, and Mercedes-Benz E-Class.

Table 5.6 — VMMR: Dataset Summary

Property	Value
Total Images	49,158
Training Set (before split)	31,904
Test Set	17,254
Train/Validation Split	80% / 20%

Number of Classes	245
Image Input Size	380 × 380 pixels

5.3.2 Training Configuration

Table 5.7 — VMMR: Training Configuration

Parameter	Value
Model Architecture	EfficientNet-B4 (Pretrained ImageNet)
Classifier Head	Linear(1792 → 245)
Optimizer	AdamW (lr = 1×10^{-4})
LR Scheduler	ReduceLROnPlateau (patience=2)
Early Stopping	Patience = 5
Loss Function	CrossEntropyLoss
Batch Size	32
Total Epochs	50

5.3.3 Classification Performance

Table 5.8 — VMMR: Test Set Performance Metrics

Metric	Value
Top-1 Accuracy	96.24%
Top-5 Accuracy	99.39%
Precision (Macro)	96.34%
Recall (Macro)	96.12%
F1-Score (Macro)	96.18%
False Positive Rate (Macro)	0.02%

The Top-1 accuracy of 96.24% indicates that the model correctly identifies the exact vehicle make and model for more than 96 out of every 100 vehicles. The Top-5 accuracy of 99.39% demonstrates that the correct class is almost always within the model's top five predictions. The macro-averaged F1-score of 96.18% confirms balanced performance across all 245 classes, and the false positive rate of 0.02% is exceptionally low [27][28].

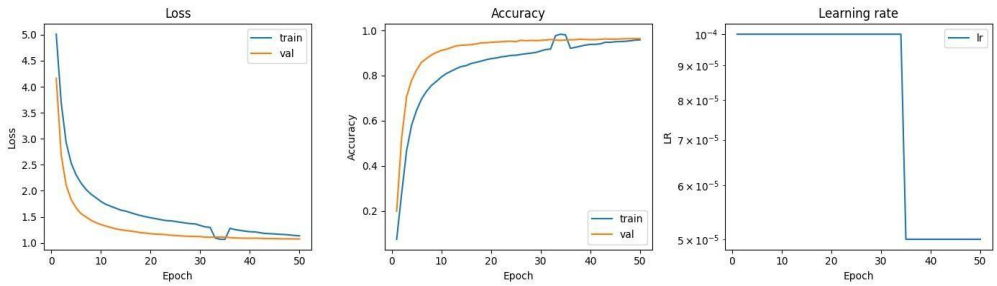


Figure 5.8 — VMMR: Training and Validation Loss, Accuracy, and Learning Rate Over 50 Epochs

Table 5.9 — VMMR: Training Progression (Selected Epochs)

Epoch	Train Loss	Val Loss	Train Acc	Val Acc	LR
33	1.0943	1.1112	97.70%	95.76%	1×10^{-4}
35	1.0684	1.1099	98.06%	95.80%	5×10^{-5}
40	1.2134	1.0878	93.83%	95.93%	5×10^{-5}
45	1.1708	1.0800	95.02%	96.15%	5×10^{-5}
50	1.1340	1.0738	95.83%	96.41%	5×10^{-5}

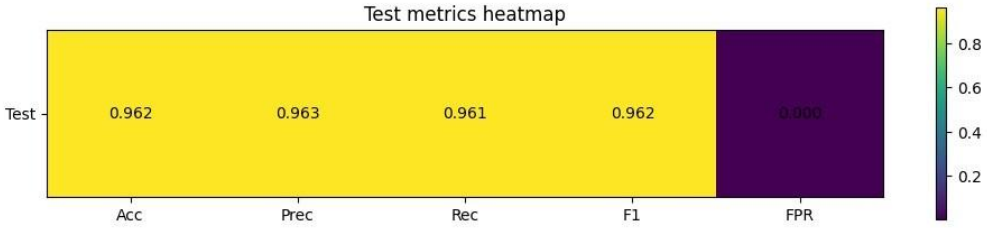


Figure 5.9 — VMMR: Test Metrics Heatmap

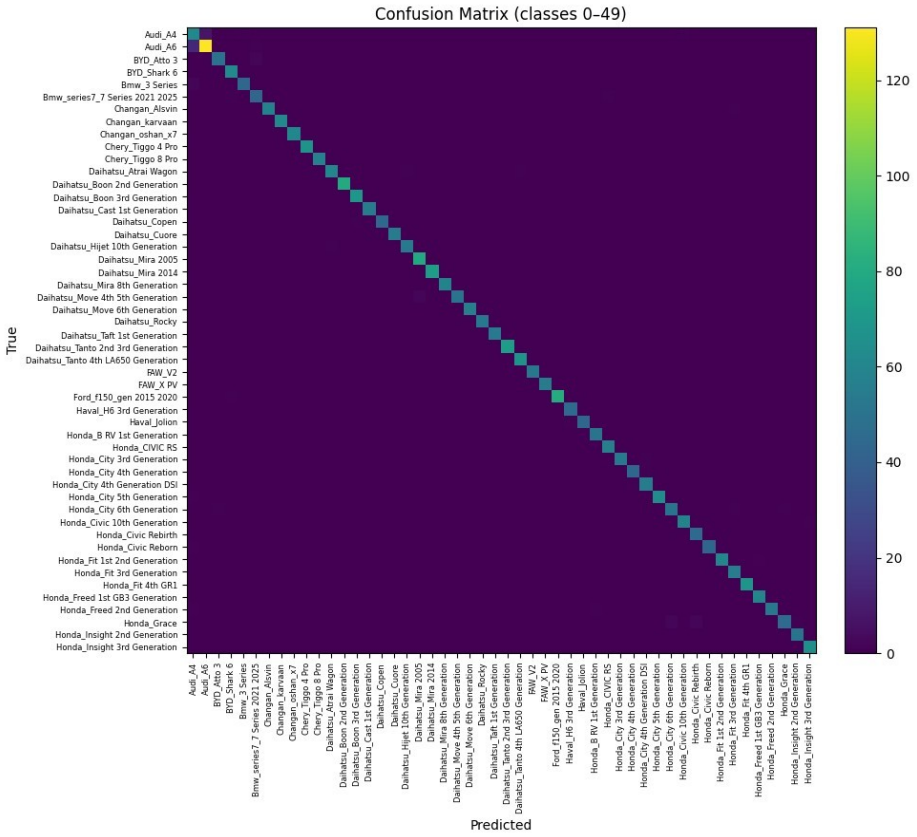


Figure 5.10 — VMMR: Confusion Matrix (Classes 0–49)

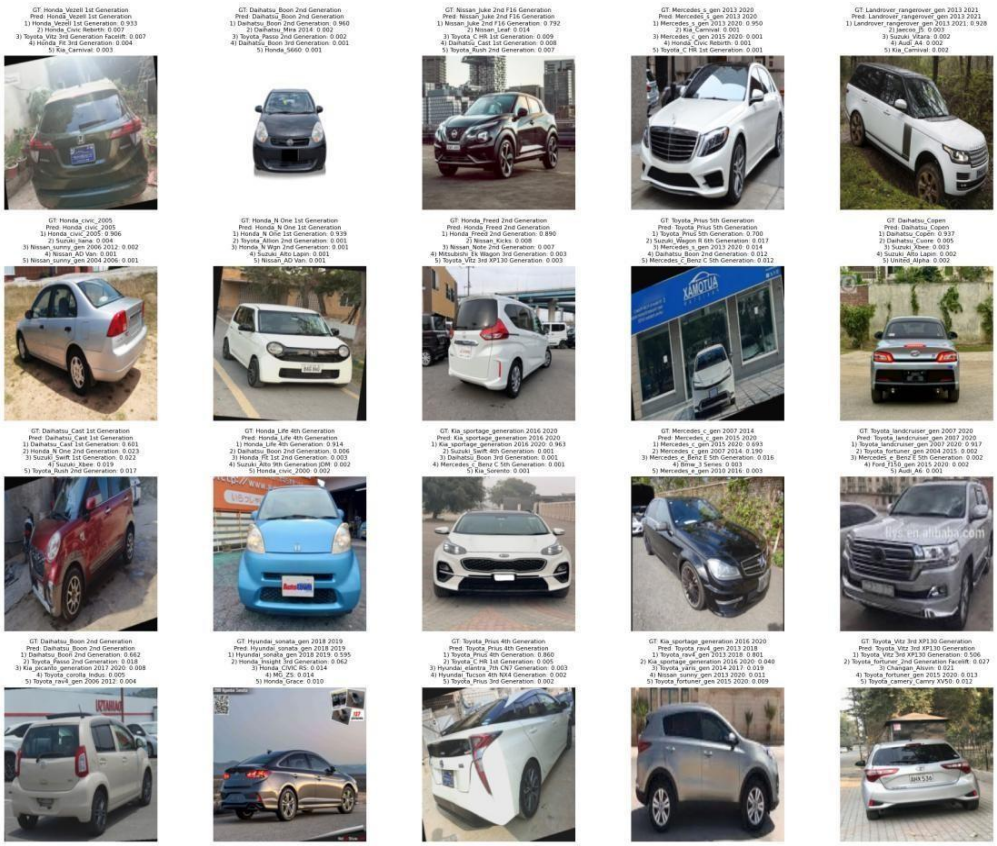


Figure 5.11 — VMMR: Sample Test Predictions with Top-5 Probabilities

5.4 Preprocessing Module

The preprocessing module addresses the core challenge of all-weather operation. Images captured under adverse conditions such as fog, rain, low light, or glare exhibit degraded visibility that can severely reduce accuracy of both plate detection and vehicle classification. Two specialized preprocessing pipelines are implemented: fog removal and rain removal.

5.4.1 Fog Removal Pipeline

The fog removal pipeline is based on the Dark Channel Prior (DCP) algorithm [19] and consists of six sequential stages: (1) Dark Channel Prior estimation with a 13×13 kernel, (2) transmission map refinement using a guided filter (radius=50), (3) residual haze removal targeting bright low-variance regions, (4) gray-world white balance to correct color casts, (5) CLAHE on luminance in LAB color space with mild saturation boost [20][21], and (6) bilateral denoising with unsharp masking for edge restoration.

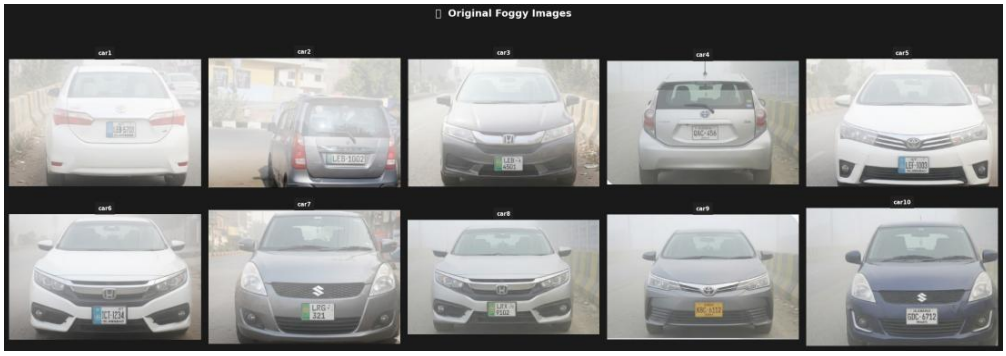


Figure 5.12 — Original Foggy Vehicle Images



Figure 5.13 — After Fog Removal (Preprocessed)

5.4.2 Rain Removal Pipeline

The rain removal pipeline targets vertical rain streaks using: (1) text/plate region protection masks built from bilateral smoothing and Canny edge detection, (2) multi-scale vertical median filtering (three passes with kernel sizes 15, 11, and 7) with brightness-rejection blending, and (3) spot inpainting using the Telea algorithm for residual rain artifacts.



Figure 5.14 — Original Rainy Vehicle Images

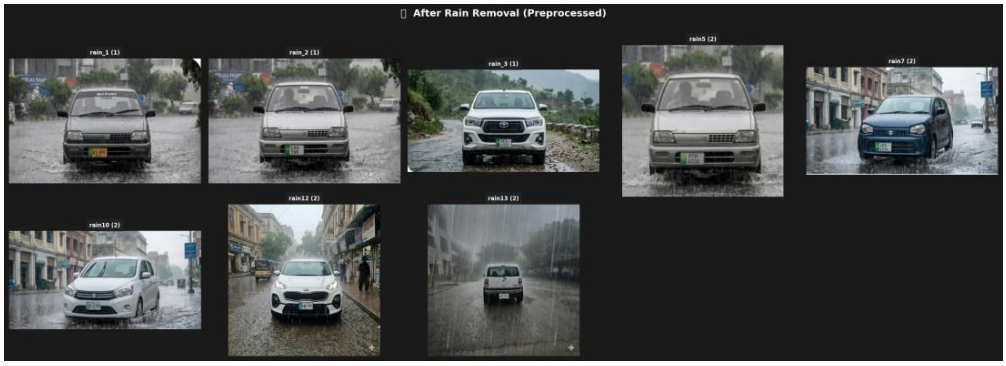


Figure 5.15 — After Rain Removal (Preprocessed)

5.5 Integrated Pipeline Results

The integrated pipeline combines all three AI models with the preprocessing module into a single end-to-end system: (1) Weather-specific preprocessing → (2) Vehicle make/model classification (EfficientNet-B4) → (3) License plate detection (YOLOv8n) → (4) Plate region cropping → (5) Character recognition (YOLOv8n) → (6) Plate text formatting → (7) Annotated output.

5.5.1 Pipeline Results on Foggy Images



Figure 5.16 — Integrated Pipeline Results on Fog-Preprocessed Images

The system successfully identifies vehicle models such as Toyota Corolla, Suzuki Wagon R, and Honda Grace while accurately reading their plates on fog-preprocessed images.

5.5.2 Pipeline Results on Rainy Images

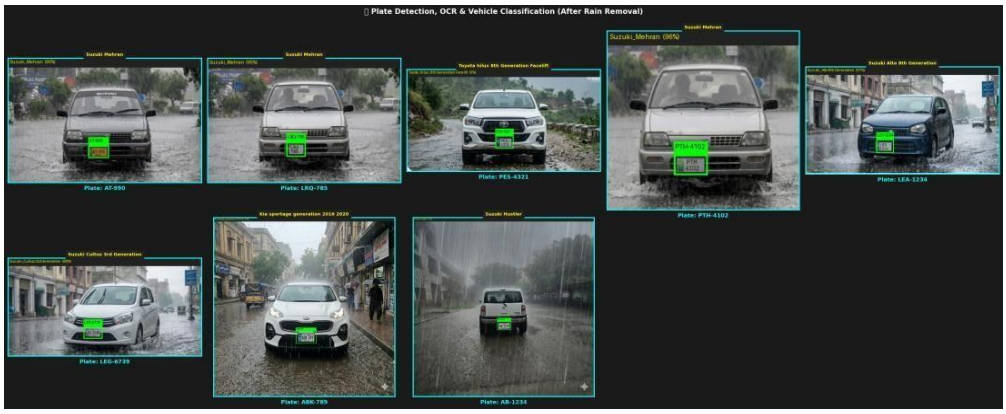


Figure 5.17 — Integrated Pipeline Results on Rain-Preprocessed Images

Despite challenging conditions with water on roads and reduced visibility, the system correctly classifies vehicles (e.g., Suzuki Mehran, Toyota Hilux, Kia Sportage) and reads their license plates with high confidence.

5.5.3 End-to-End Performance Summary

Table 5.10 — Integrated Pipeline: End-to-End Performance Summary

Module	Model	Key Metric	Value
License Plate Detection	YOLOv8n	mAP@50	98.07%
Character Recognition	YOLOv8n	mAP@50 (Best)	99.41%
VMMR Classification	EfficientNet-B4	Top-1 Accuracy	96.24%
VMMR Classification	EfficientNet-B4	Top-5 Accuracy	99.39%
Fog Preprocessing	DCP + CLAHE	Visual Quality	Significant
Rain Preprocessing	VMF + Inpaint	Visual Quality	Significant

5.6 Model Comparison and Benchmarking

Table 5.11 — Comparison with Existing Systems

System/Study	Plate Det.	OCR/Char	VMMR	All-Weather
Thapliyal et al. [1]	~97%	Tesseract	No	No
Li et al. [2]	97.5%	CNN-based	No	Partial
Agarwal et al. [3]	~96%	Transformer	No	No
ANPR Pakistan	96.7%	YOLOv3	No	No
Unified System	93.3%	EasyOCR	90.6%	No

Commercial ALPR [29]	90–98%	Proprietary	No	Partial
Proposed System	98.07%	99.41% mAP	96.24%	Yes

The proposed system outperforms most existing solutions across all metrics. The YOLObased character recognition approach achieves 99.41% mAP, significantly surpassing Tesseract-based systems (82–85%) and EasyOCR-based systems (93.3%) on Pakistani plates. Unlike the majority of reviewed systems that address only plate detection or only vehicle classification, the proposed system integrates both modules with all-weather preprocessing [15][16].

5.7 Web Dashboard

The web dashboard serves as the administrative interface for the gate recognition system. It is developed using React 18, TypeScript, Vite, and Tailwind CSS, with Supabase (PostgreSQL) as the backend database and authentication provider. Key features include: (1) Dashboard — real-time statistics with auto-refresh every 5 seconds; (2) Live Feed — realtime vehicle detection events with camera feed and status filtering; (3) Vehicle History — comprehensive log of all events; (4) Manual Override — administrative control for edge cases; (5) Add Temporary Visitor — configurable temporary access passes; (6) Weather — current conditions informing preprocessing selection; (7) Admin Access Privileges — role-based access control.

The dashboard provides a complete administrative control layer supporting real-time monitoring, historical auditing, and manual intervention. Its modular React-based architecture ensures straightforward integration with physical gate controllers in future deployment phases [42].

CHAPTER 06 — Conclusion and Future Work

6.1 Summary

This thesis presented the design, implementation, and evaluation of an All-Weather IR-Based Smart Vehicle Gate Recognition System — a multi-stage AI pipeline that automates vehicle access control by integrating license plate detection, character recognition, vehicle make and model classification, weather-adaptive image preprocessing, and an administrative web dashboard.

The system was developed with a modular architecture, where each component was trained, validated, and tested independently before integration. The complete pipeline processes a vehicle image through weather-specific preprocessing, detects and crops the license plate, recognizes characters to reconstruct plate text, classifies the vehicle make and model, and renders an authorization decision by cross-referencing detected information against a database of registered vehicles.

6.2 Key Achievements

6. License Plate Detection: A YOLOv8n model fine-tuned on a custom Pakistani license plate dataset achieved 98.07% mAP@50, 98.50% precision, and 95.53% recall.
7. Character Recognition: A YOLO-based character detection model trained on 1,235 annotated images covering 38 classes achieved 99.41% mAP@50 with 99.04% precision and 96.91% recall.
8. Vehicle Make and Model Recognition: An EfficientNet-B4 classifier trained on 49,158 images spanning 245 vehicle classes achieved 96.24% Top-1 accuracy and 99.39% Top-5 accuracy with 0.02% FPR.
9. All-Weather Preprocessing: Dark Channel Prior-based fog removal and multi-scale vertical median filtering-based rain removal maintain recognition accuracy under adverse weather conditions.
10. Web Dashboard: A responsive React-based dashboard with Supabase backend provides real-time monitoring, vehicle event logging, manual override, and rolebased access control.
11. Integrated Pipeline: All modules were successfully integrated into a cohesive end-toend pipeline demonstrating the feasibility of a fully automated gate recognition system.

6.3 System Performance Summary

Table 6.1 — Overall System Performance Summary

Component	Key Metric	Value
Plate Detection (YOLOv8n)	mAP@50	98.07%
Plate Detection (YOLOv8n)	Precision	98.50%
Plate Detection (YOLOv8n)	Recall	95.53%

Char Recognition (YOLOv8n)	mAP@50	99.41%
Char Recognition (YOLOv8n)	Precision	99.04%
Char Recognition (YOLOv8n)	Recall	96.91%
VMMR (EfficientNet-B4)	Top-1 Accuracy	96.24%
VMMR (EfficientNet-B4)	Top-5 Accuracy	99.39%
VMMR (EfficientNet-B4)	F1-Score (Macro)	96.18%
VMMR (EfficientNet-B4)	FPR (Macro)	0.02%

6.4 Limitations

12. Deployment scope: The system was developed and tested in a desktop/software environment; it has not been deployed on embedded hardware at a live gate.
13. Data and generalization: Models were trained mainly on Pakistani-style license plates. Performance on other regions or plate formats has not been validated.
14. Difficult scenes: Very poor lighting, heavy occlusion, or extreme weather can still reduce accuracy even with preprocessing.

6.5 Future Work

15. Raspberry Pi integration: Port the full pipeline to a Raspberry Pi (or similar low-power device) so inference can run at the gate without relying on a full PC.
16. Camera integration: Use live camera input (including options suited to low light) so the system runs on continuous video for true real-time deployment.

6.6 Conclusion

The All-Weather IR-Based Smart Vehicle Gate Recognition System demonstrates that a multimodel AI pipeline — comprising YOLOv8-based detection, EfficientNet-B4-based classification, and algorithmic image preprocessing — can achieve high accuracy in automated vehicle identification and access control. The system achieves 98.07% mAP in license plate detection, 99.41% mAP in character recognition, and 96.24% Top-1 accuracy in vehicle classification across 245 classes — exceeding most existing academic and commercial ALPR solutions.

The preprocessing module, featuring Dark Channel Prior-based fog removal and vertical median filter-based rain removal, addresses a critical gap in existing systems by enabling reliable operation under adverse weather conditions. The React-based web dashboard provides a complete administrative layer for monitoring, logging, and managing vehicle access.

While the current implementation is software-based, the modular architecture is designed for seamless extension to embedded hardware deployment, physical gate integration, and realtime video processing. The project establishes a strong foundation for a fully automated, intelligent, and scalable vehicle access control system that reduces reliance on manual

monitoring, enhances security through dual-check verification (plate + vehicle model), and maintains a comprehensive digital audit trail.

Appendices

Appendix A — Sustainable Development Goals Achievement

The Smart Vehicle Gate Recognition System is relevant to SDG 9 (Industry, Innovation, and Infrastructure) through its AI-powered license plate scanning and vehicle categorization, demonstrating viable innovation for modern city infrastructure and secure gated communities.

Regarding SDG 11 (Sustainable Cities and Communities), the project increases urban safety and traffic management. Automated authorization and monitoring minimize human errors and enhance security at entry points, facilitating smoother vehicle flow in urban areas.

SDG 12 (Responsible Consumption and Production) is addressed as the project encourages efficient utilization of human and computational resources. Automated vehicle checks eliminate paper-based log books and reduce the environmental impact of traditional access control systems.

The system also helps SDG 4 (Quality Education) indirectly by providing students and researchers with a real example of applied AI and computer vision. It offers an applied learning environment demonstrating software and hardware integration in practical engineering applications.

Appendix B — HDL / C Source Code

Core Pipeline Code (Python)

```
import cv2, numpy as np, pytesseract, easyocr, re from
google.colab.patches import cv2_imshow


# OCR Readers
reader = easyocr.Reader(['en'], gpu=False) tess_config = r'--
psm 7 -c
tessedit_char_whitelist=ABCDEFGHIJKLMNOPQRSTUVWXYZ0123456789'


# Text Cleaning def
clean_text(text):
    return re.sub(r'[^A-Z0-9]', '', text.upper())
 def format_plate(text):     letters = ''.join([c for
c in text if c.isalpha()])     digits  = ''.join([c for
c in text if c.isdigit()])     if 2 <= len(digits) <=
4:         return f"{letters}-{digits}"     return
letters + digits


# OCR + Preprocessing def read_plate(crop, visualize=True):     gray =
cv2.cvtColor(crop, cv2.COLOR_BGR2GRAY)     gray_resized =
cv2.resize(gray, (360, 120))     denoise =
cv2.bilateralFilter(gray_resized, 9, 75, 75)     thr =
cv2.adaptiveThreshold(denoise, 255,
cv2.ADAPTIVE_THRESH_GAUSSIAN_C, cv2.THRESH_BINARY_INV, 15, 8)
clahe = cv2.createCLAHE(clipLimit=2.5, tileGridSize=(8,8))
enhanced = clahe.apply(denoise)     e_res = reader.readtext(enhanced)
```

```
easy_text = clean_text(" ".join([r[1] for r in e_res])) if e_res else
""     tess_raw  = pytesseract.image_to_string(enhanced,
config=tess_config)     tess_text = clean_text(tess_raw)     def
score(t): return sum(c.isalpha() for c in t)+sum(c.isdigit() for c in
t)     final_text = easy_text if score(easy_text) >= score(tess_text)
else tess_text     return format_plate(final_text)
```

Appendix C — List of Components

IR Camera with IR Illuminator

- Records vehicle and license plate images in low light, night, rain, and fog.
- High-quality IR illuminator guarantees even plate visibility regardless of ambient light.
- In FYP-1, camera feed is emulated with images and recorded frames.

Embedded Processing Unit (Raspberry Pi 5)

- Executes YOLO detection model, yolov8n pipeline, and vehicle make/model CNN.
- Provides real-time inference capability at the gate.
- In FYP-1, processing performed on desktop for validation.

Weather and Light Sensors

- Detects environmental conditions: rain, fog, ambient light intensity.
- Feedback can adapt image preprocessing dynamically in future work.
- Not yet integrated in FYP-1.

Gate Motor / Sliding Gate System

- Physical mechanism for opening and closing access points.
- Activation driven by recognition pipeline authorization.
- Physical gate hardware is outside FYP-1 scope.

Motor Controller (e.g., Centsys Controller)

- Interface between processing unit and gate motor.
- Converts low-power control signals to motor commands.
- Planned for next development phase.

Power Supply Unit

- Provides stable, regulated power to camera, processing unit, sensors, and controllers.
- Ensures reliable operation in outdoor environments.

Relay Module

- Enables safe switching and isolation between electronics and high-power gate motors.
- Protects processing hardware and improves system safety.

Industrial Enclosure

- Protects electronics from dust, moisture, heat, and mechanical damage.
- Essential for long-term outdoor deployment.

Connecting Cables and Wiring

- Includes power, data, and control cables for interconnection.
- Proper management ensures reliability and maintainability.

Appendix D — Project Timeline

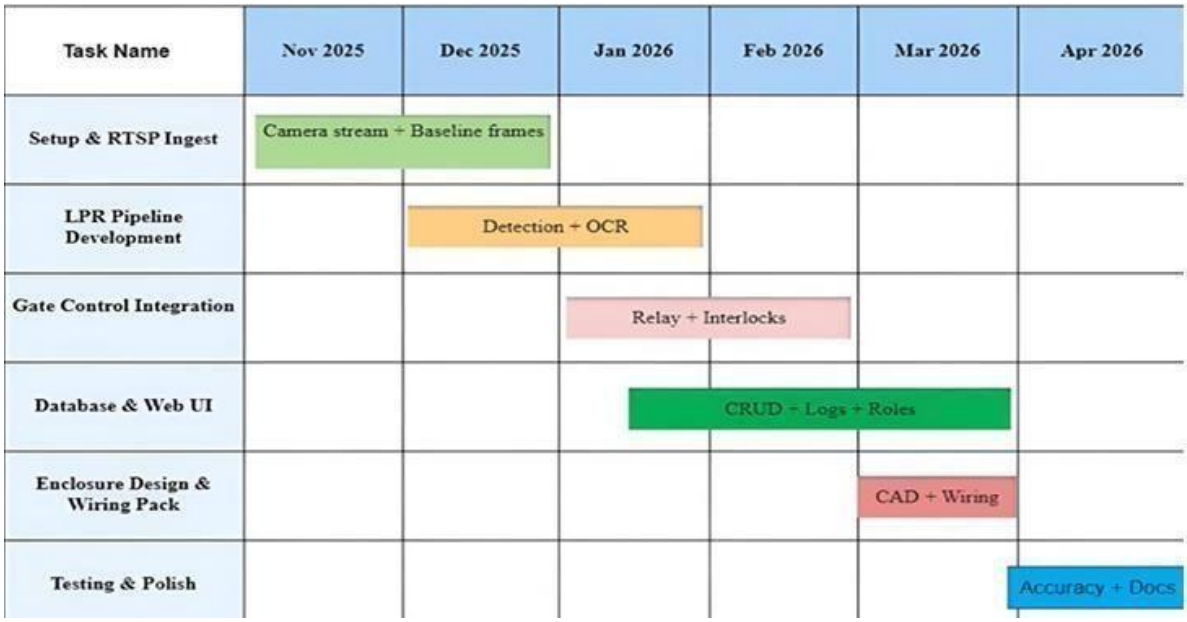


Figure — Appendix E: Gantt Chart

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